

Experiment Results — Symbolically Bounded Latent Recursion

Introduction

This document tracks the incremental experiment log for **Symbolically Bounded Latent Recursion** — a neuro-symbolic architecture that combines a small transformer (the Tiny Recursive Model, or TRM) with a Z3 SMT solver to solve constraint satisfaction problems like Sudoku.

The goal is to build a model that reasons iteratively: update a latent state across multiple recursion steps, refining its predictions the way a human would — check the grid, find violations, correct them, repeat. Pure neural recursion should, in theory, do this naturally. In practice, a critical failure mode called **representation collapse** causes the latent state to explode exponentially, destroying the recursive signal within a few steps. The hypothesis is that injecting symbolic constraints from Z3 into the recursive loop can bound the latent trajectory to a valid manifold, preventing collapse and enabling the model to reason deeper than it could alone.

The path follows three phases:

- 1. **Baseline characterization** (Runs 001–005) — Establish how the TRM behaves without any symbolic intervention. Quantify the collapse rate, identify where the architecture breaks, and find a task scale (4×4 Sudoku) where learning is possible despite collapse.
- 2. **Symbolic integration at the output layer** (Runs 006–007) — Apply Z3 constraints to the model's predictions (logit pruning). Test whether penalizing invalid outputs is sufficient to improve accuracy. This is the simplest integration point.
- 3. **Architectural stabilization and latent-space integration** (Run 008+) — Fix collapse at the source by normalizing the latent state inside the recurrence. Then apply Z3 constraints within the stabilized loop itself — not at the output, but at the representation level. This is the core thesis: symbolic bounds in latent space should enable recursive reasoning that neither pure neural nor output-level symbolic approaches can achieve alone.

Each run entry below records the hypothesis, configuration, results, diagnostics, and interpretation. The summary sections at the end aggregate patterns, track open questions, and prioritize next experiments.

Sudoku Baseline Runs (2026-04-03)

These initial experiments establish the pure neural baseline for the TRM on 9×9 Sudoku (no symbolic bounds). The goal is to quantify representation collapse and motivate the Z3 integration.

Run 001 — sudoku_20260403_111010

Hypothesis: Verify the baseline architecture trains and produces non-trivial loss curves.

Parameter	Value
dim	128
n_layers	2
n_heads	8
n_recursions	4
n_cycles	3
n_supervision	7
epochs	10
lr	3e-4
weight_decay	0.1
train / val	10,000 / 5,000
givens range	17–35

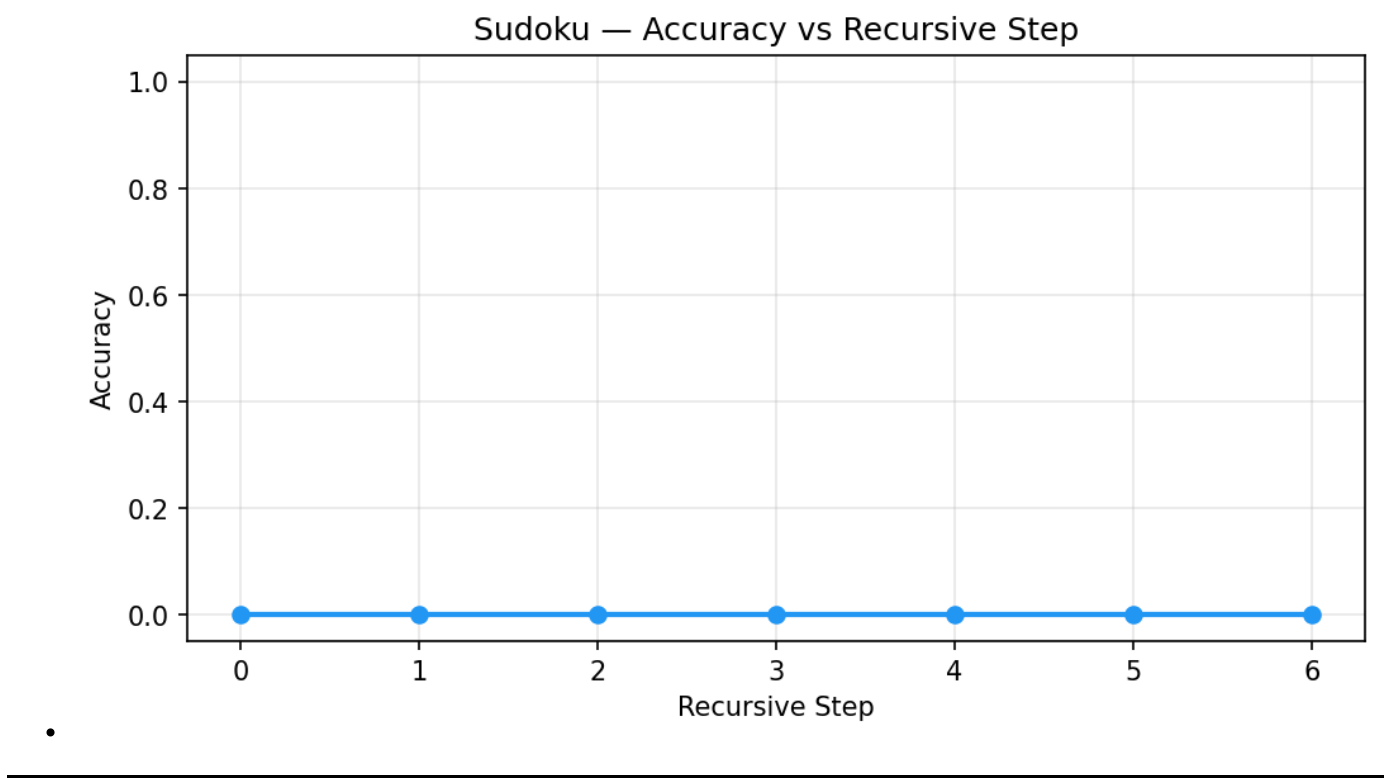
Parameter	Value
training time	~22 min

Results:

Metric	Value
Cell accuracy	20.5%
Puzzle accuracy	0.0%
Constraint satisfaction	0.0%
Best val accuracy	0.0%
Per-step accuracy (steps 0–6)	0.0% across all

Observations: Loss decreased but the model never broke past random-guess cell accuracy (~11% = 1/9). No per-step accuracy at any recursion depth. No diagnostics were captured for this run.

Plots:



Run 002 — sudoku_20260403_155953

Hypothesis: Same config as Run 001, but with the updated training loop that captures diagnostics. Confirm reproducibility and examine latent state behavior.

Parameter	Value
(same as Run 001)	
training time	~22 min

Results:

Metric	Value
Cell accuracy	19.7%
Puzzle accuracy	0.0%

Metric	Value
Constraint satisfaction	0.0%
Best val accuracy	0.0%
Final epoch loss	1.236

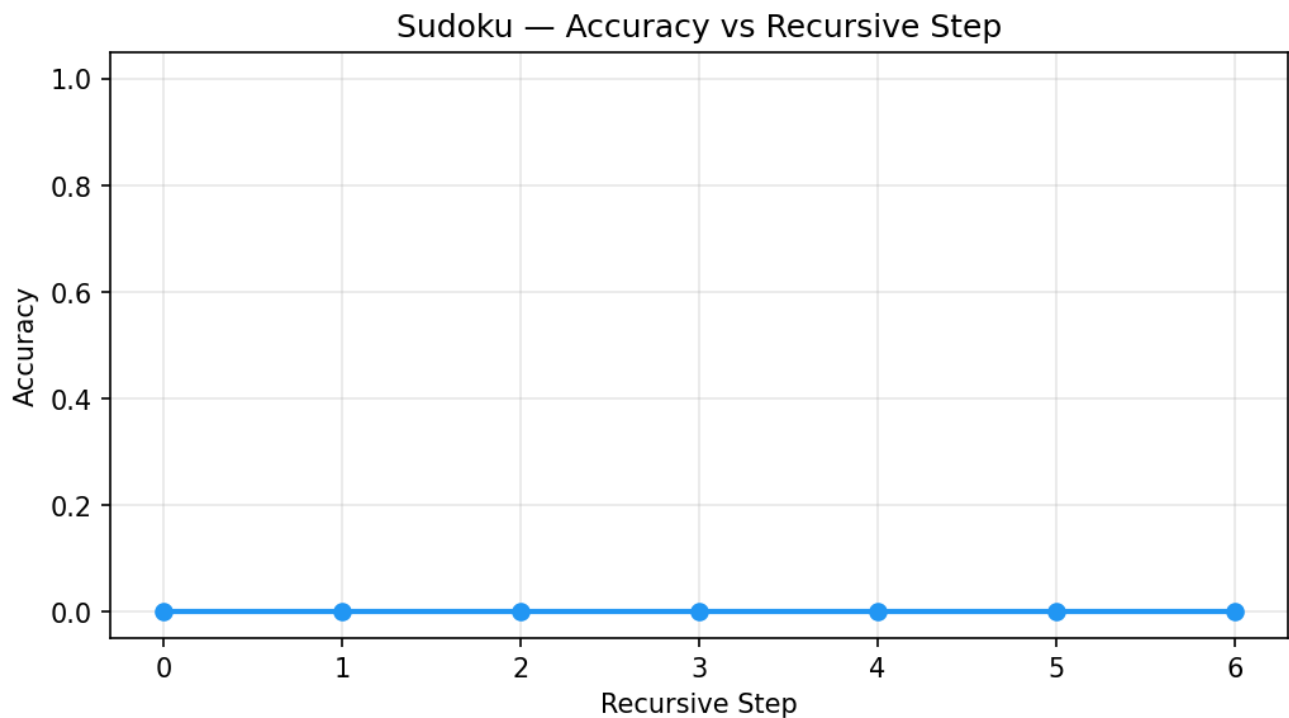
Loss curve (10 epochs): 2.506 → 2.209 → 2.195 → 2.046 → 1.765 → 1.547 → 1.374 → 1.302 → 1.265 → 1.236

Diagnostics — representation collapse confirmed:

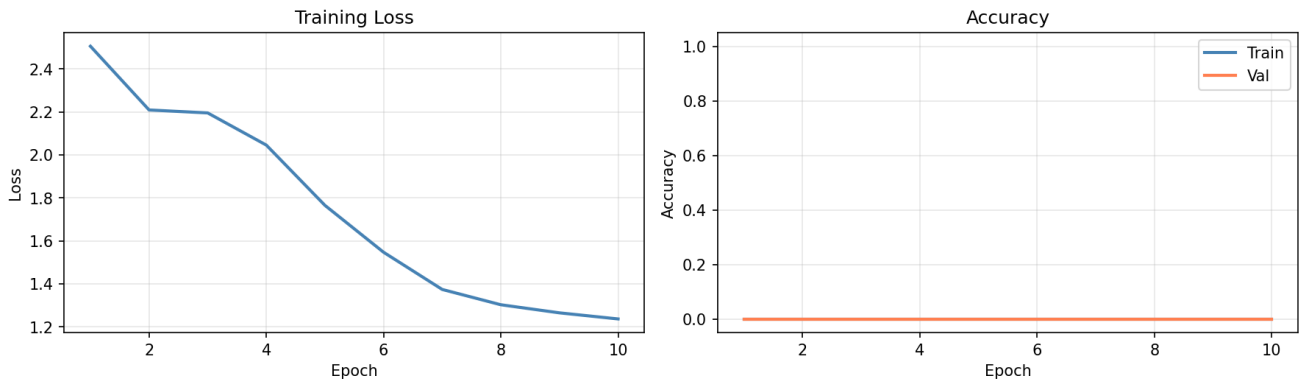
Step	z norm	y norm	z cosine	halt prob
0	2,088	2,510	—	3.9e-5
1	413,835	499,532	1.000	4.0e-5
2	81.9M	98.9M	1.000	4.0e-5
3	16.2B	19.6B	1.000	4.0e-5
4	3.21T	3.88T	1.000	4.0e-5
5	636T	768T	1.000	4.0e-5
6	1.26e17	1.52e17	1.000	4.0e-5

The z and y norms explode by ~200× per recursion step. Cosine similarity locks at 1.0 after step 1 — the direction is frozen while the magnitude blows up. The halt gate never fires (stuck near 0). This is textbook representation collapse: the recursion amplifies a fixed direction without refining content.

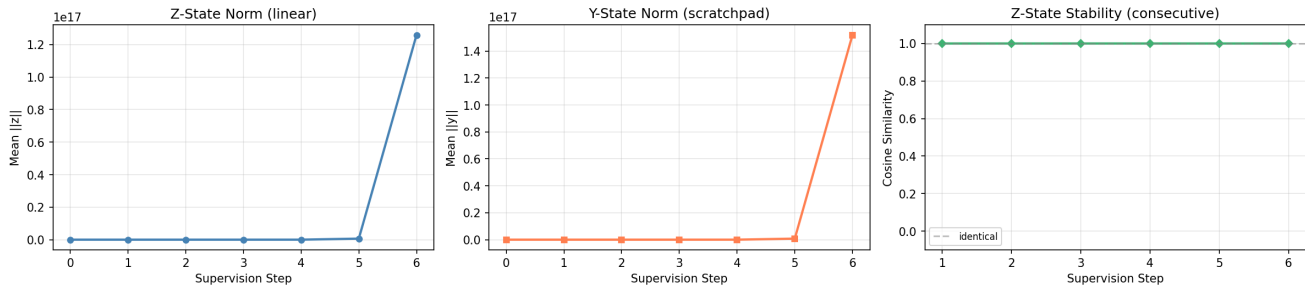
Plots:



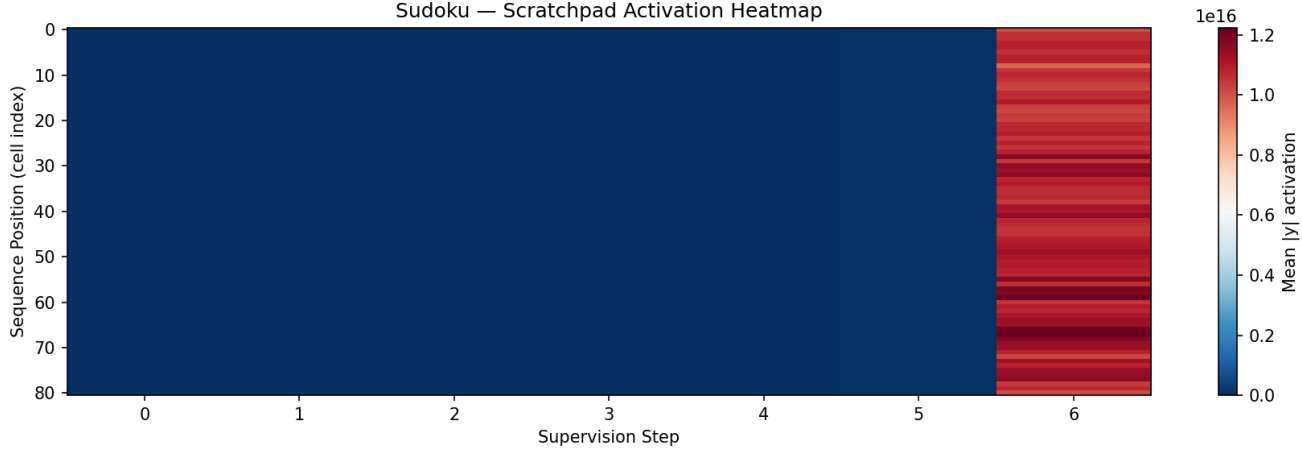
Sudoku — Training Curves



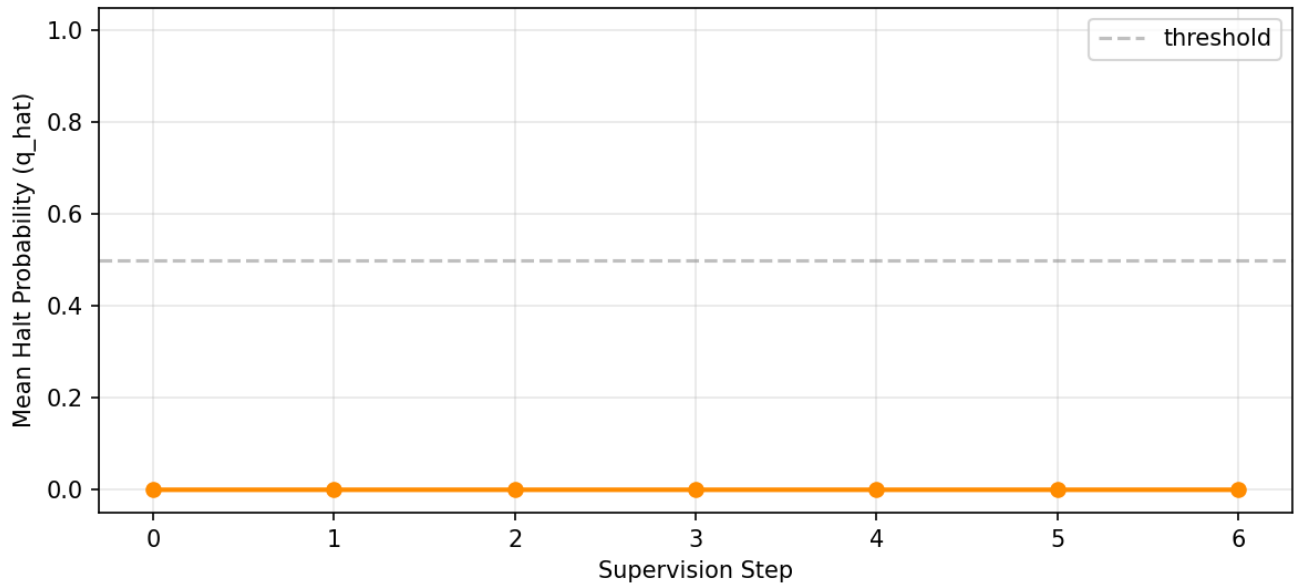
Sudoku — Z-State Evolution



Sudoku — Scratchpad Activation Heatmap



Sudoku — Halt Confidence



Hypothesis: Doubling recursions (4 → 8) and epochs (10 → 20) to see if more compute helps, or if collapse gets worse.

Parameter	Value
n_recursions	8 (was 4)
num_epochs	20 (was 10)
training time	~76 min
(all other params same)	

Results:

Metric	Value
Cell accuracy	11.1%
Puzzle accuracy	0.0%
Constraint satisfaction	0.0%
Best val accuracy	0.0%
Final epoch loss	1.081

Loss curve (20 epochs): 2.413 → ... → 1.081 (steadily decreasing, minor train acc blips of 0.01% at epochs 17–19)

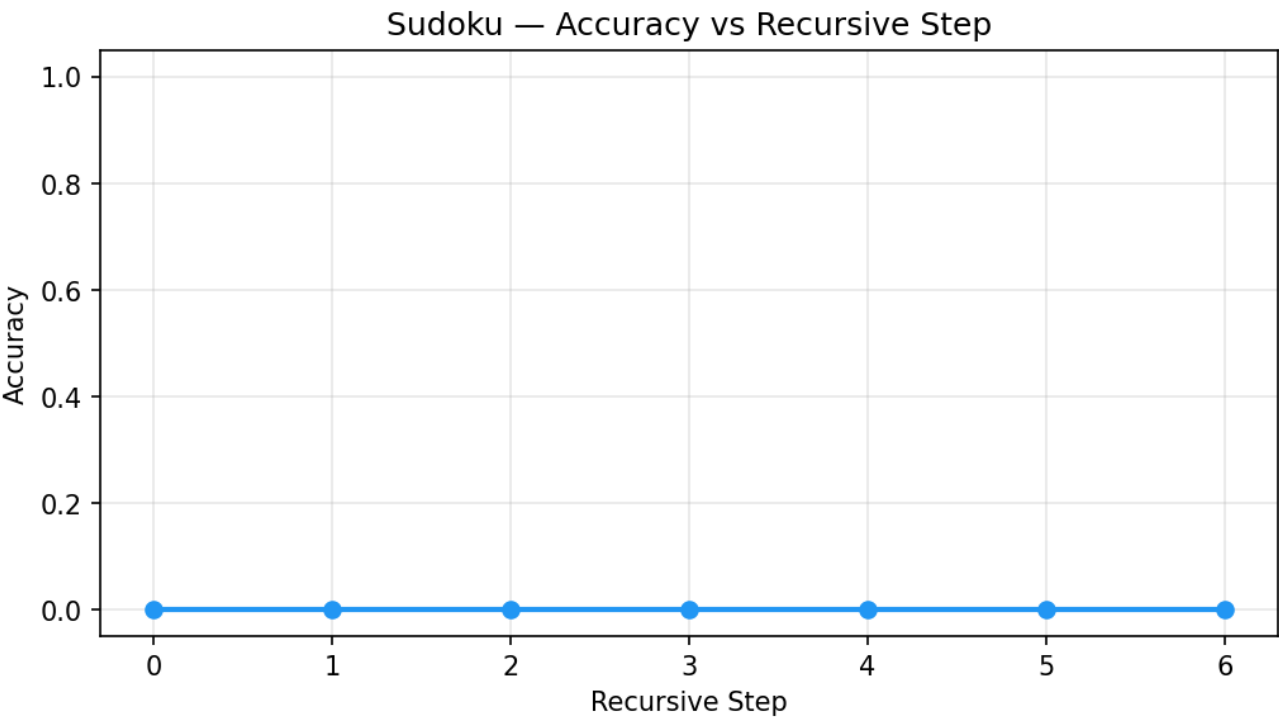
Diagnostics — collapse is catastrophically worse:

Step	z norm	y norm	z cosine	halt prob
0	8,743	9,716	—	5.3e-5
1	8.48M	9.44M	1.000	5.3e-5
2	8.23B	9.15B	1.000	5.3e-5
3	7.98T	8.88T	1.000	5.3e-5
4	7.74e15	8.61e15	1.000	5.3e-5
5	7.51e18	8.35e18	0.0	5.3e-5
6	∞	∞	0.0	0.500

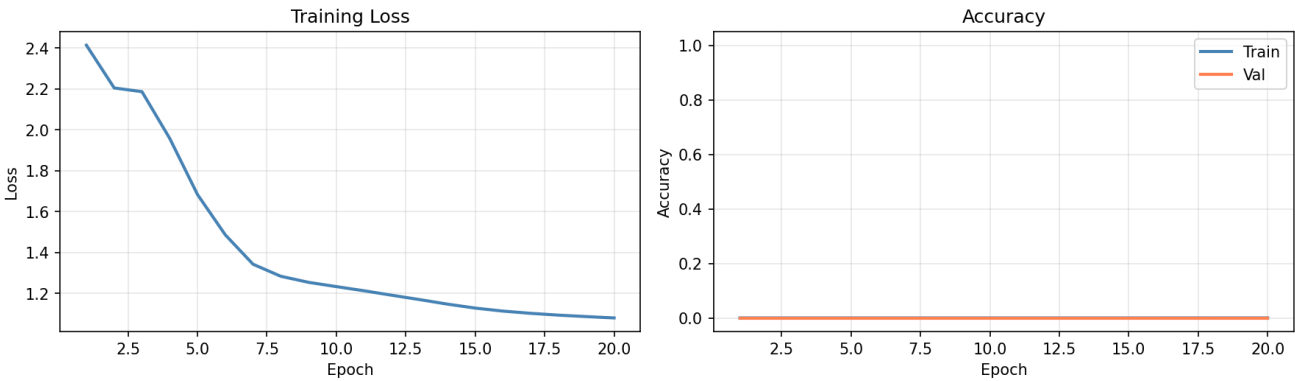
Norms hit infinity by step 6. The ~970× per-step growth rate is far worse than Run 002's ~200×. Cosine drops to 0.0 at steps 5–6 (NaN/Inf corruption). The halt gate finally fires at 0.5 on the last step — but only because the signal is numerically garbage. Cell accuracy dropped to 11.1% (random chance for 1/9), worse than the 4-recursion runs.

Key finding: More recursions without stabilization actively degrades performance. The model is strictly worse with 8 recursions than with 4.

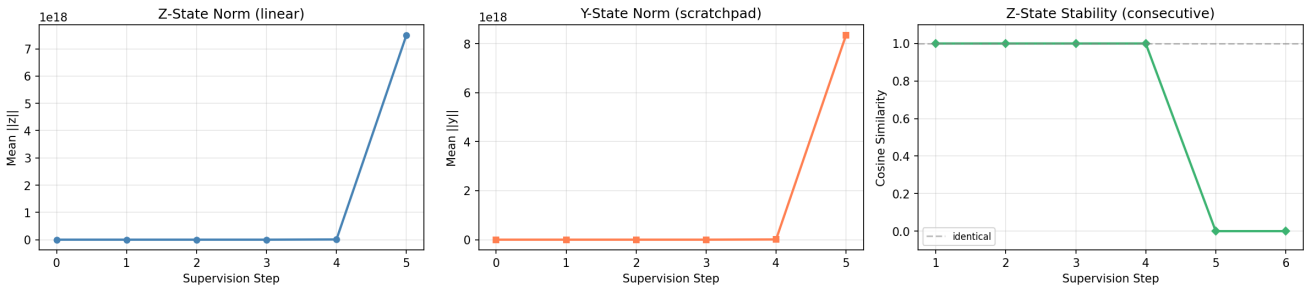
Plots:



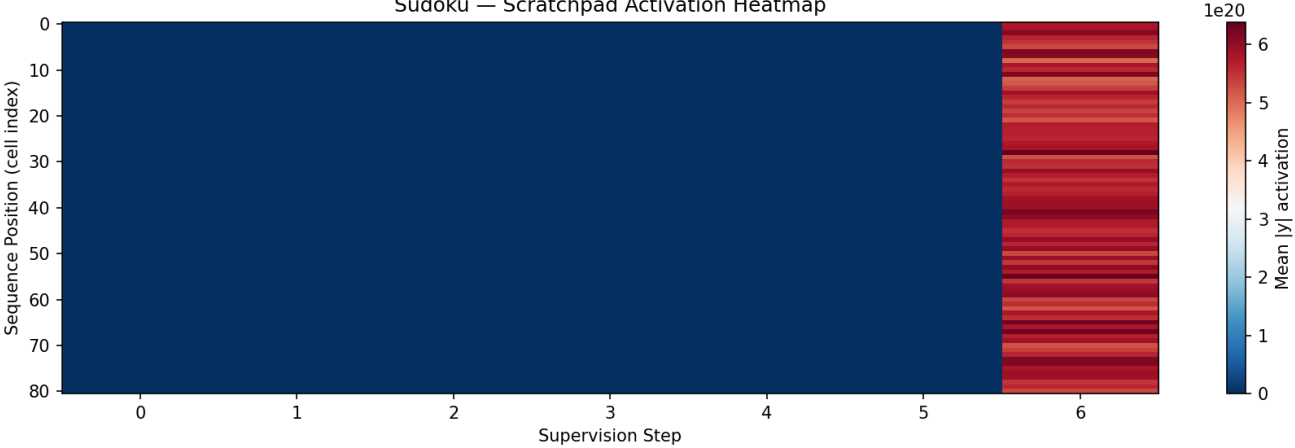
Sudoku — Training Curves

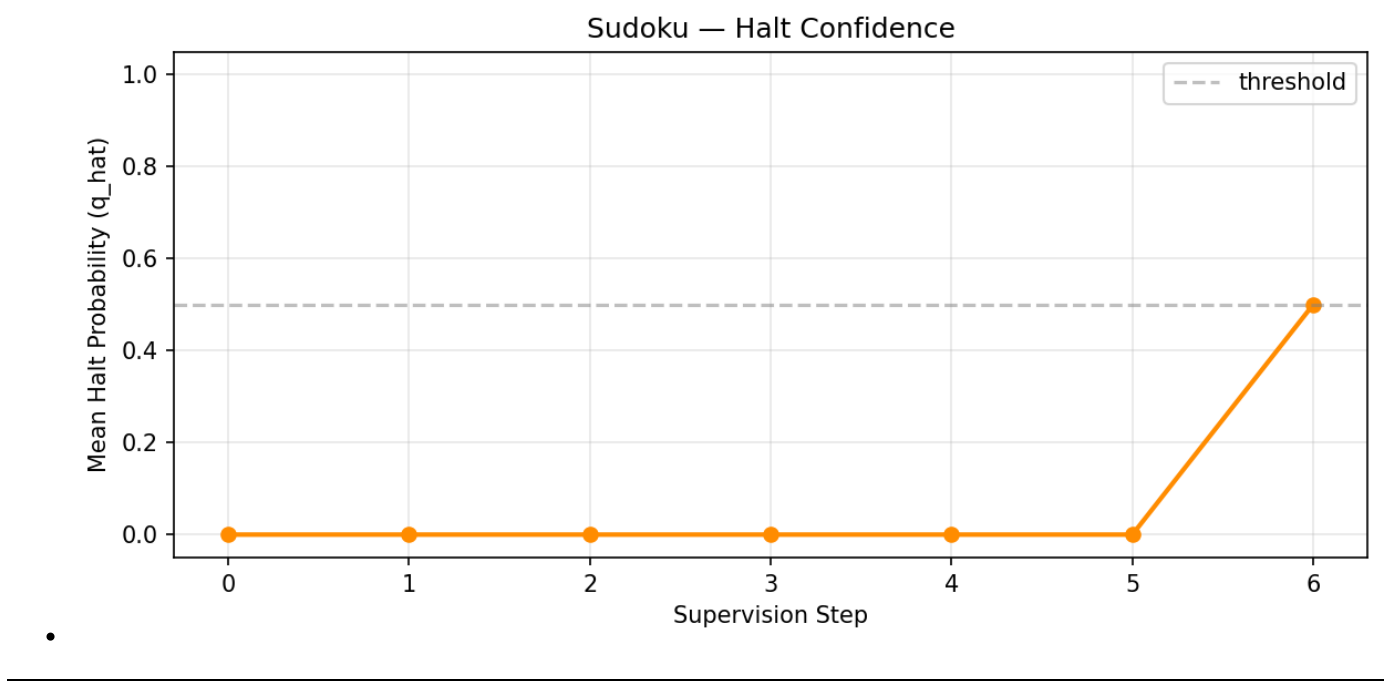


Sudoku — Z-State Evolution



Sudoku — Scratchpad Activation Heatmap





Run 004 — sudoku_20260403_215600

Hypothesis: Keep recursions at 4 but double the supervision depth (7 → 14) with 20 epochs. If the model gets gradient signal at more intermediate steps, maybe it can learn to control the recursion better.

Parameter	Value
n_recursions	4 (same as Run 002)
n_supervision	14 (was 7)
num_epochs	20 (was 10)
training time	~95 min

(all other params same)

Results:

Metric	Value
Cell accuracy	11.1%
Puzzle accuracy	0.0%
Constraint satisfaction	0.0%
Best val accuracy	0.0%
Final epoch loss	1.060

Loss curve (20 epochs): 2.418 → 2.205 → 2.186 → 1.965 → 1.710 → 1.468 → 1.316 → 1.254 → 1.218 → 1.192 → 1.169 → 1.151 → 1.133 → 1.118 → 1.103 → 1.091 → 1.080 → 1.073 → 1.065 → 1.060

Loss converges slightly lower than Run 003 (1.060 vs 1.081). Faint train accuracy blips at epochs 17/19 (0.02–0.03%), but val accuracy remains 0.0% throughout.

Diagnostics — deeper supervision does not prevent collapse:

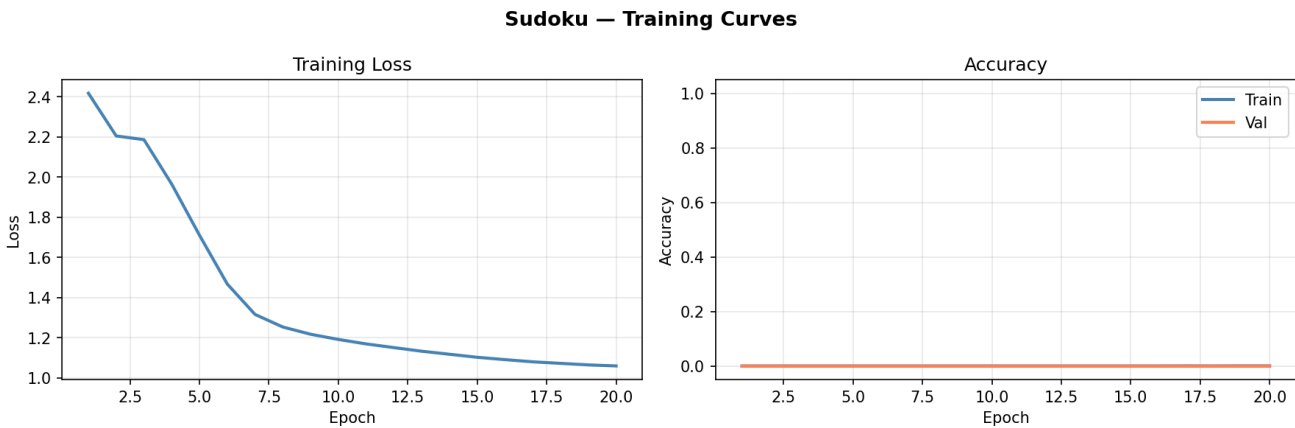
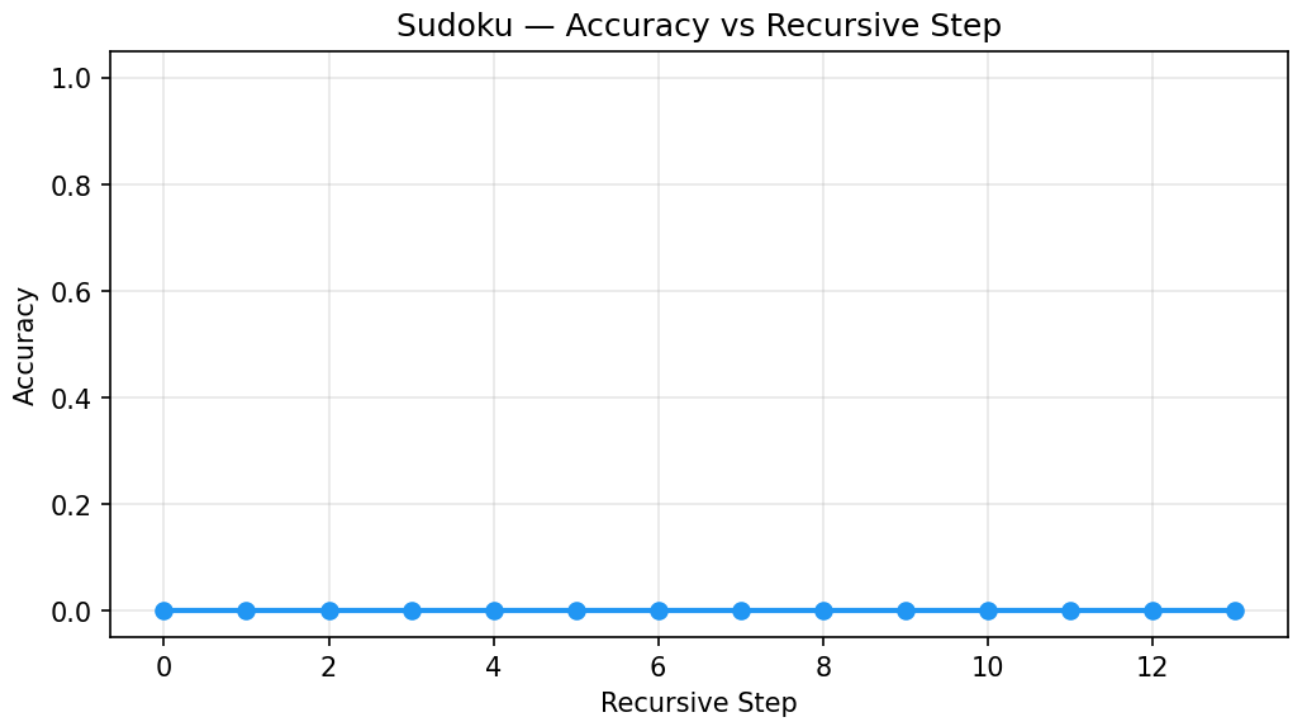
Step	z norm	y norm	z cosine	halt prob
0	1,692	2,034	—	4.0e-6
1	335,830	405,374	1.000	4.0e-6
2	66.5M	80.3M	1.000	4.0e-6

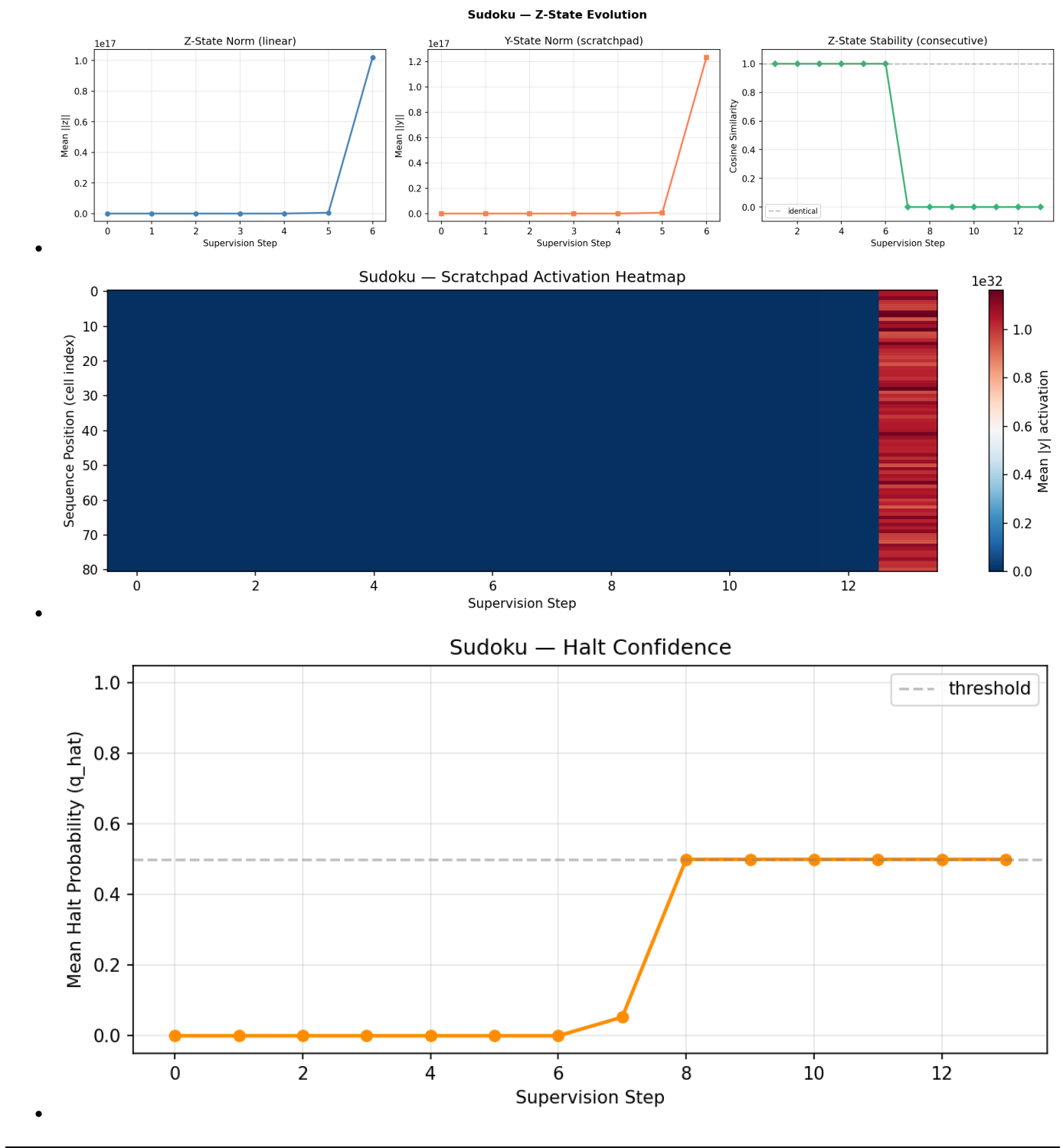
Step	z norm	y norm	z cosine	halt prob
3	13.2B	15.9B	1.000	4.0e-6
4	2.61T	3.15T	1.000	4.0e-6
5	516T	623T	1.000	4.0e-6
6	1.02e17	1.23e17	1.000	4.0e-6
7-13	∞	∞	0.0	0.053 $\rightarrow$ 0.500

The per-step growth rate (~198x) is nearly identical to Run 002, confirming this is an intrinsic property of the architecture at 4 recursions. The key difference is that with 14 supervision steps (vs 7), the model now has 7 additional steps where it's reading out predictions from infinitely-blown latent states. Norms overflow to ∞ by step 7. The halt gate is even more suppressed than before (4.0e-6 vs 4.0e-5 in Run 002) during the valid steps, then snaps to 0.5 once the signal is corrupted.

Key finding: Doubling supervision depth without addressing the underlying norm explosion just gives the model more steps of garbage to read from. Cell accuracy regressed from ~20% (Runs 001-002) to 11.1% (random chance) — the same degradation seen in Run 003 with 8 recursions. The additional supervision taps at infinity-corrupted steps may actually poison the gradient signal, pulling the model toward worse overall representations.

Plots:





4x4 Mini-Sudoku Experiments (2026-04-04)

These experiments switch to 4x4 mini-Sudoku to reduce task difficulty and introduce Z3 symbolic bounds. They reveal the **Readout Trap** — the central paradox motivating latent-space symbolic projection.

Run 005 — `sudoku_20260404_170039` — 4x4 Pure Neural Baseline

Hypothesis: Switch from 9x9 to 4x4 Sudoku to reduce task difficulty. If 9x9 cell accuracy was capped at ~20% (Runs 001–002), the simpler 4x4 task (4 digits, 16 cells, 2x2 blocks) should be learnable even with representation collapse, establishing the pure-neural ceiling before introducing Z3.

Parameter	Value
dim	128

Parameter	Value
n_layers	2
n_heads	8
n_recursions	4
n_cycles	3
n_supervision	7
epochs	20
lr	3e-4
weight_decay	0.1
grid_size	4x4 (was 9x9)
use_z3_pruning	False
train / val	5,000 / 1,000
min_givens / max_givens	4 / 10
training time	~10 min (589 s)
device	cuda

Results:

Metric	Value
Cell accuracy	80.0% (random chance = 25%)
Puzzle accuracy	38.7%
Constraint satisfaction	38.8%
Best val accuracy	38.7% (epoch 20 — still climbing)

Loss curve (20 epochs): 1.647 → 1.457 → 1.384 → 1.296 → 1.036 → 0.807 → 0.644 → 0.566 → 0.524 → 0.498 → 0.474 → 0.448 → 0.423 → 0.390 → 0.369 → 0.357 → 0.337 → 0.321 → 0.310 → 0.303

Val accuracy curve (20 epochs):

0→0→0→0→0→0→0→0→0→0→0.1%→0.1%→0.2%→0.9%→1.7%→4.6%→8.8%→13.7%→20.2%→29.4%→38.7%

Loss decreases steadily and has not plateaued. Val accuracy is on a steep upward trajectory at epoch 20 — this run was still improving when it ended. Training accuracy reached ~6.5% by epoch 14.

Diagnostics — collapse still present but the network learns despite it:

Step	z norm	y norm	z cosine	halt prob
0	1,902	2,284	—	0.088
1	376,823	454,853	1.000	0.089
2	74.6M	90.1M	1.000	0.089
3	14.8B	17.8B	1.000	0.089
4	2.92T	3.53T	1.000	0.089
5	579T	699T	1.000	0.089
6	1.15e17	1.38e17	1.000	0.089

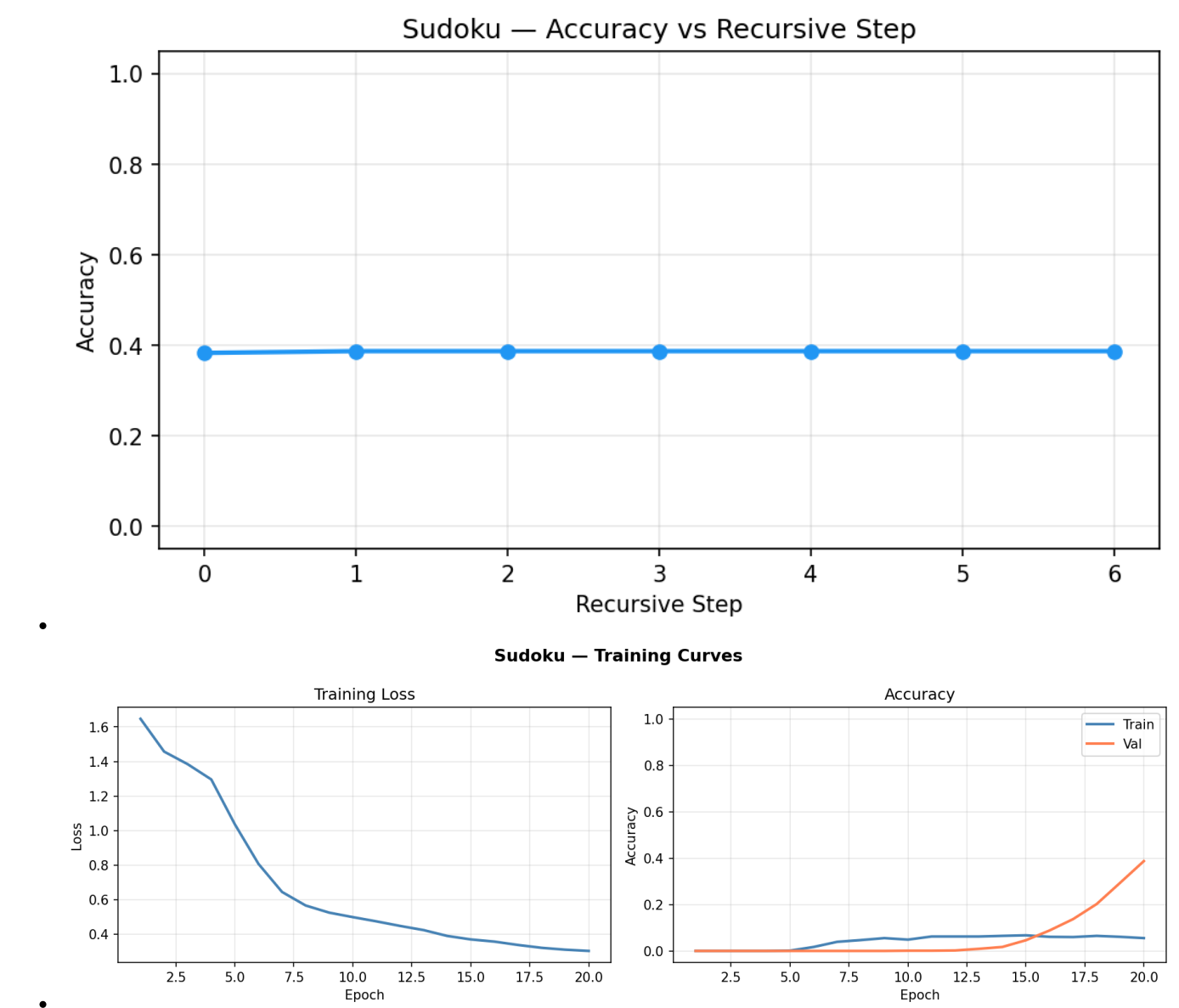
Per-step growth rate ~198x — identical to all other 4-recursion runs. Cosine locks at 1.0 after step 0. The halt gate is the most active of any run so far at 0.088–0.089, though still non-functional. Despite representation collapse, the network achieves 38.7% puzzle accuracy — it has learned to encode valid 4x4 solutions into the first-step readout before the latent state explodes.

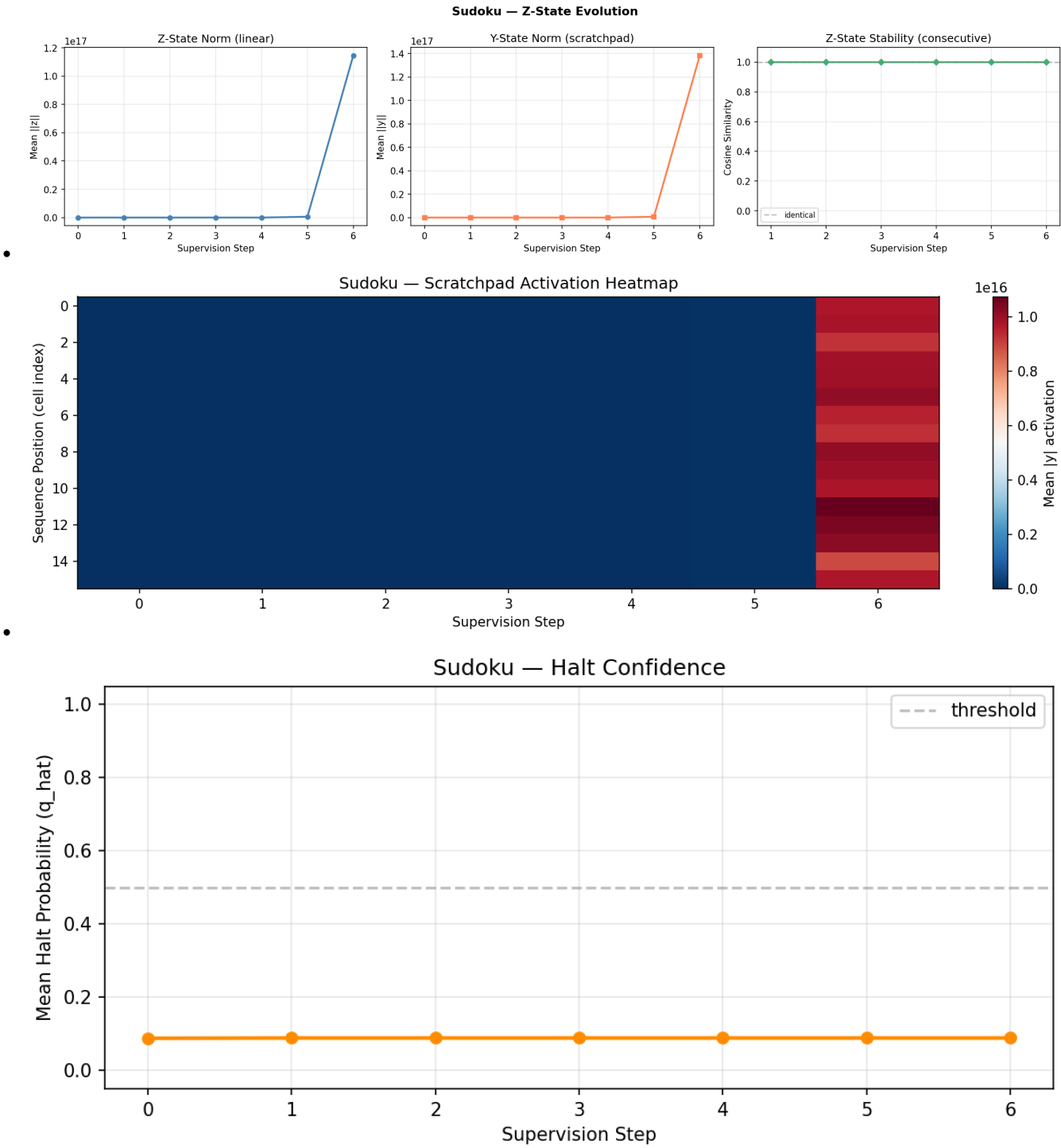
Per-step accuracy: 38.3% at step 0, 38.7% at steps 1–6. Near-identical across all steps, confirming the frozen-direction collapse: each step reads from the same latent direction at different magnitudes.

Key findings:

- 1. **4×4 is learnable even with full representation collapse.** The pure neural network achieves 38.7% puzzle accuracy and 80.0% cell accuracy — dramatically better than any 9×9 run. The simpler task lets the model encode enough structure in the first recursion step's readout to solve nearly 4 in 10 puzzles.
- 2. **Val accuracy was still climbing at epoch 20.** The steep trajectory (0% → 38.7% in the final 10 epochs) suggests the model had not converged. More epochs could push accuracy significantly higher. This stands in stark contrast to the 9×9 runs where accuracy was permanently stuck at 0%.
- 3. **This is the critical baseline for evaluating Z3 integration.** Any Z3-enhanced run on 4×4 must beat 38.7% puzzle accuracy to demonstrate that symbolic bounds add value beyond what the pure neural network achieves alone.

Plots:





Run 006 — sudoku_20260404_185547 — Z3 Logit Pruning (Unscaled)

Hypothesis: Enable Z3 logit pruning on the 4×4 task. If the Z3 penalty suppresses invalid digit candidates at the output head, puzzle accuracy should exceed the pure-neural baseline of 38.7% (Run 005).

Parameter	Value
(same as Run 005 except:)	
use_z3_pruning	True (first time)
training time	~9 min (538 s)

Results:

Metric	Value
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Metric	Value
Cell accuracy	46.7% (random chance = 25%)
Puzzle accuracy	0.1%
Constraint satisfaction	0.1%
Best val accuracy	1.5% (epoch 2)

Loss curve (20 epochs): 226.8M → 211.7M → 215.2M → 220.8M → 222.9M → 222.9M → 225.2M → 226.6M → 201.9M → 201.1M → 189.9M → 178.4M → 179.0M → 179.0M → 186.6M → 199.8M → 180.7M → 181.6M → 164.9M → 163.0M

Val accuracy peaks at 1.5% in epoch 2 then degrades monotonically to 0.1% by epoch 20 — the opposite trajectory of Run 005's steady climb.

Diagnostics — Z3 does not prevent norm explosion:

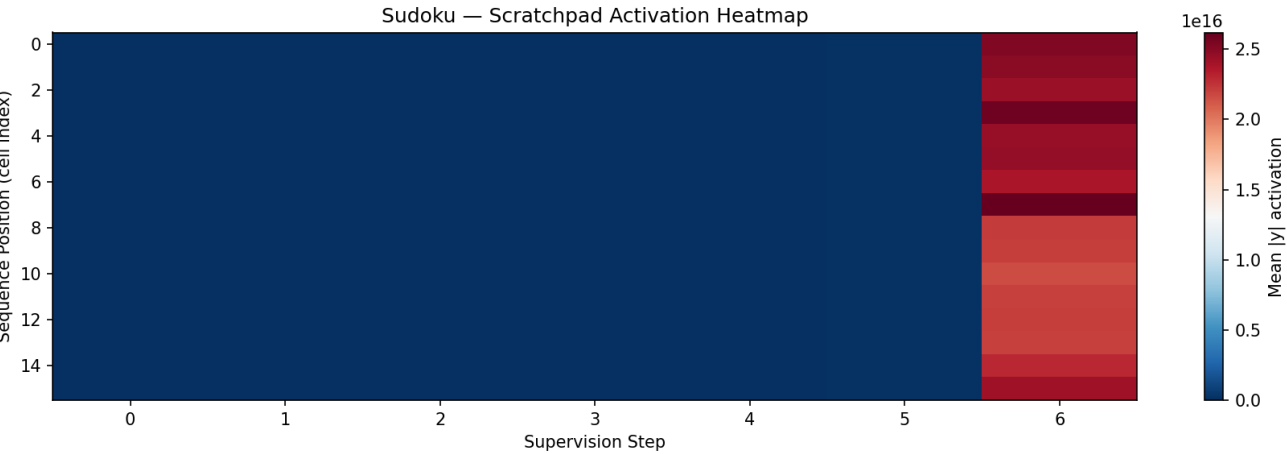
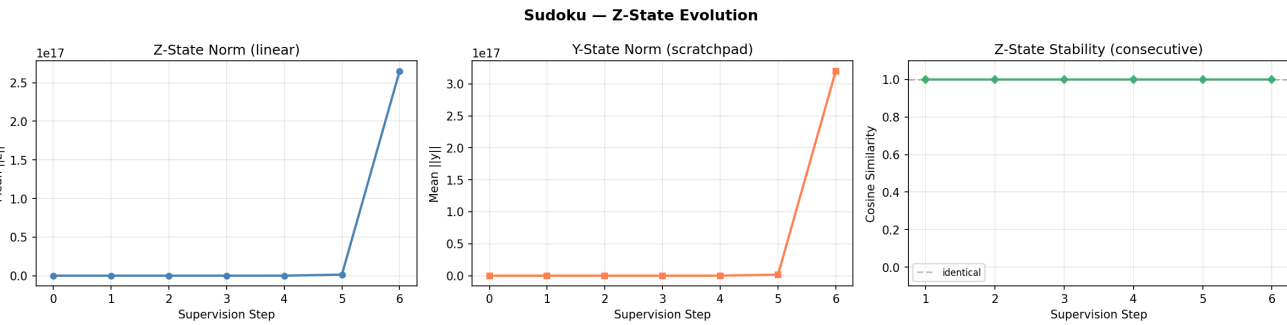
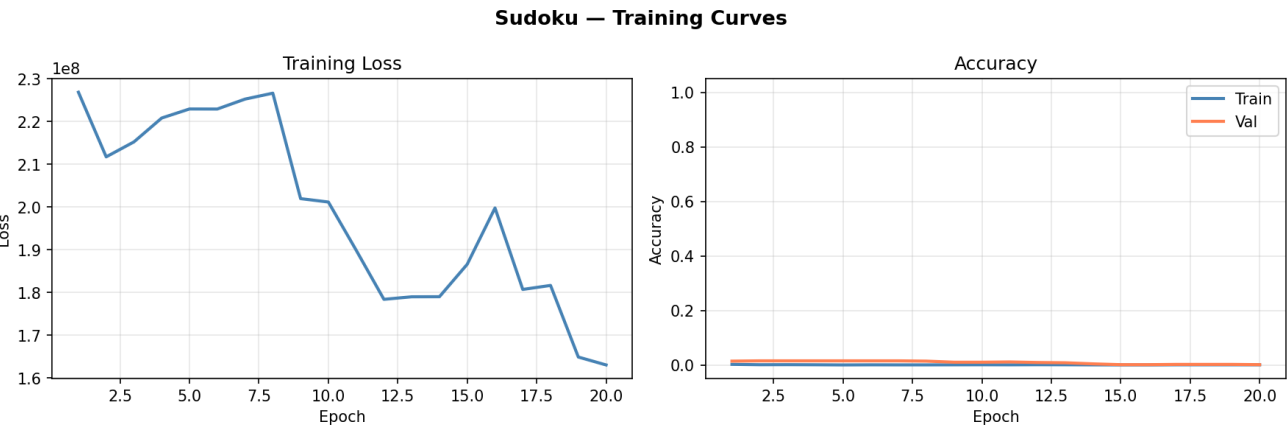
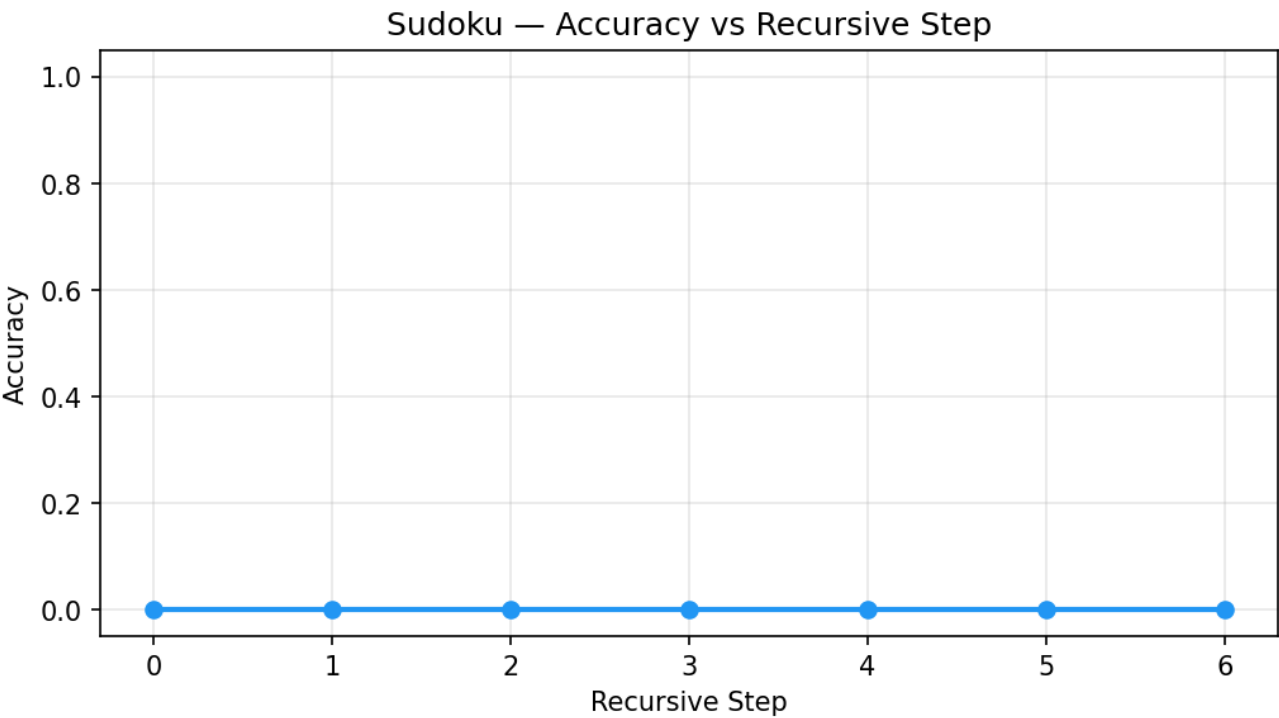
Step	z norm	y norm	z cosine	halt prob
0	4,380	5,282	—	3.6e-3
1	871,553	1.05M	1.000	3.6e-3
2	172.6M	208.3M	1.000	3.6e-3
3	34.2B	41.2B	1.000	3.6e-3
4	6.76T	8.17T	1.000	3.6e-3
5	1.34e15	1.62e15	1.000	3.6e-3
6	2.65e17	3.20e17	1.000	3.6e-3

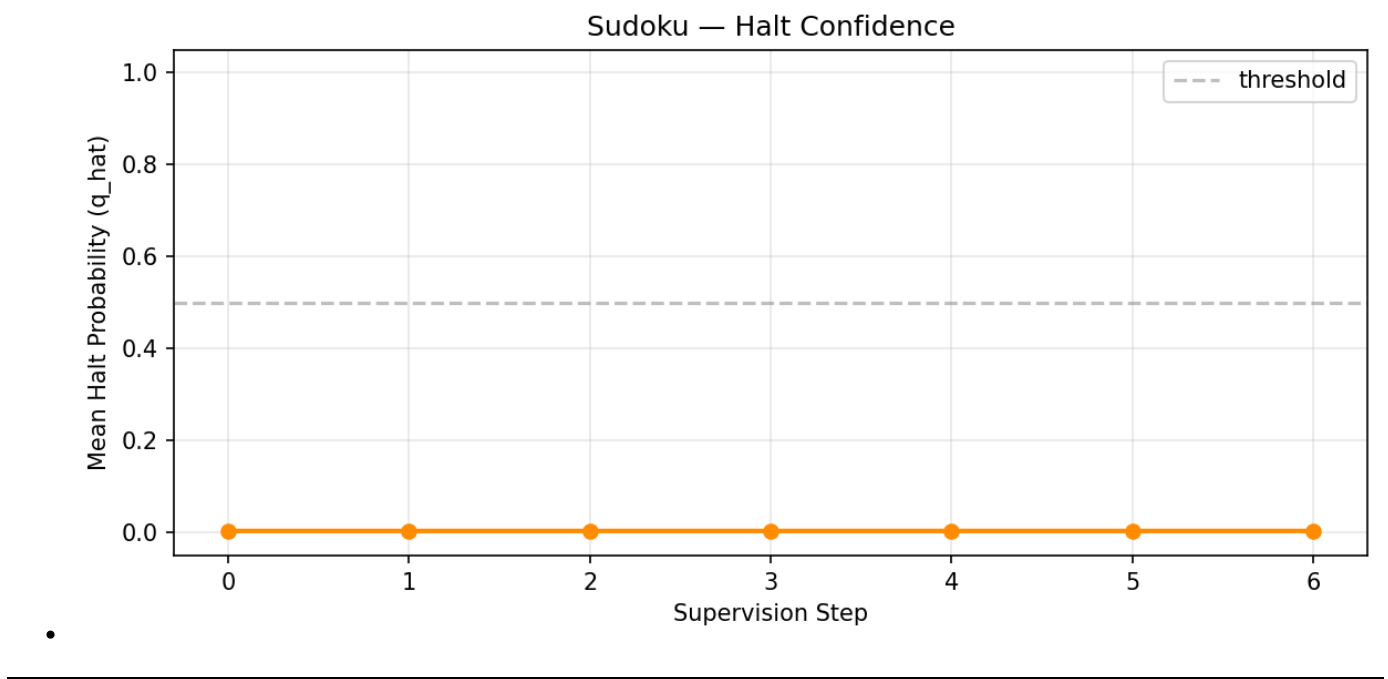
Per-step growth rate ~198× — identical to Run 005's pure-neural collapse. The Z3 penalty operates at the logit level (output head) and does not touch the latent recursion.

Key findings:

- Z3 with unscaled penalty is catastrophically worse than pure neural.** Puzzle accuracy dropped from **38.7% → 0.1%** — a 387× regression. The raw Z3 penalty (-1e9 logit masking) dominates the loss by ~8 orders of magnitude (226M vs ~0.3 CE), making the training signal almost entirely "reduce the Z3 penalty" rather than "solve Sudoku."
- The Z3 penalty actively destroyed what the pure network learned.** Run 005's pure network was climbing steadily; this run's val accuracy peaked at 1.5% in epoch 2 then collapsed to 0.1%. The unscaled penalty gradient overwhelmed the CE gradient, preventing the network from learning the pattern-memorization strategy that worked in Run 005.
- Halt gate suppressed.** Halt probability dropped from 0.088 (Run 005) to 0.0036 — the Z3-dominated gradient confused the halt mechanism.

Plots:





Run 007 — `sudoku_20260404_192851` — Z3 Logit Pruning (Scaled)

Hypothesis: Same config as Run 006 (4x4, Z3 enabled), but with the Z3 penalty scaling fix applied. If the unscaled penalty was the primary problem in Run 006 (losses in the 100M range), properly weighting Z3 against CE should recover performance. Target: beat the pure-neural baseline of 38.7% (Run 005).

Parameter	Value
(same as Run 006 except:)	
Z3 penalty	scaled (was raw -1e9)
training time	~8 min (461 s)

Results:

Metric	Value
Cell accuracy	65.7% (random chance = 25%)
Puzzle accuracy	14.5%
Constraint satisfaction	14.5%
Best val accuracy	14.7% (epoch 1)

Loss curve (20 epochs): 0.831 → 0.760 → 0.729 → 0.686 → 0.670 → 0.670 → 0.665 → 0.659 → 0.660 → 0.654 → 0.656 → 0.657 → 0.658 → 0.652 → 0.650 → 0.653 → 0.647 → 0.638 → 0.644 → 0.637

Loss is back in a sane range (~0.83 → 0.64), confirming the Z3 penalty scaling was fixed. But val accuracy is flat at 14.5–14.7% across all 20 epochs — the model converges instantly and then plateaus, in stark contrast to Run 005's steady climb.

Diagnostics — collapse unchanged, halt gate more active:

Step	z norm	y norm	z cosine	halt prob
0	3,323	4,000	—	0.083
1	659,856	796,506	1.000	0.083
2	130.7M	157.7M	1.000	0.083
3	25.9B	31.2B	1.000	0.083
4	5.12T	6.18T	1.000	0.083

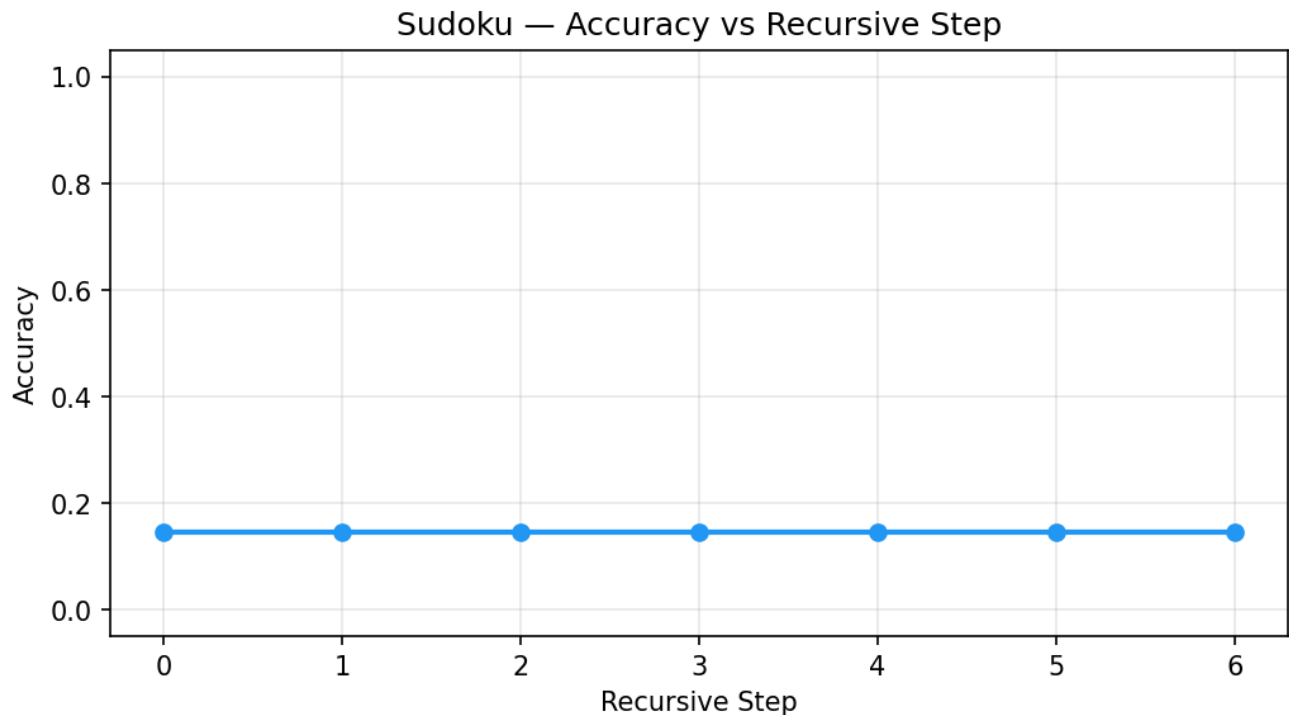
Step	z norm	y norm	z cosine	halt prob
5	1.01e15	1.22e15	1.000	0.083
6	2.01e17	2.42e17	1.000	0.083

Per-step growth rate ~198x — unchanged. Per-step accuracy: 14.5% at all 7 steps (0–6), identical across all supervision levels, confirming cosine-locked collapse.

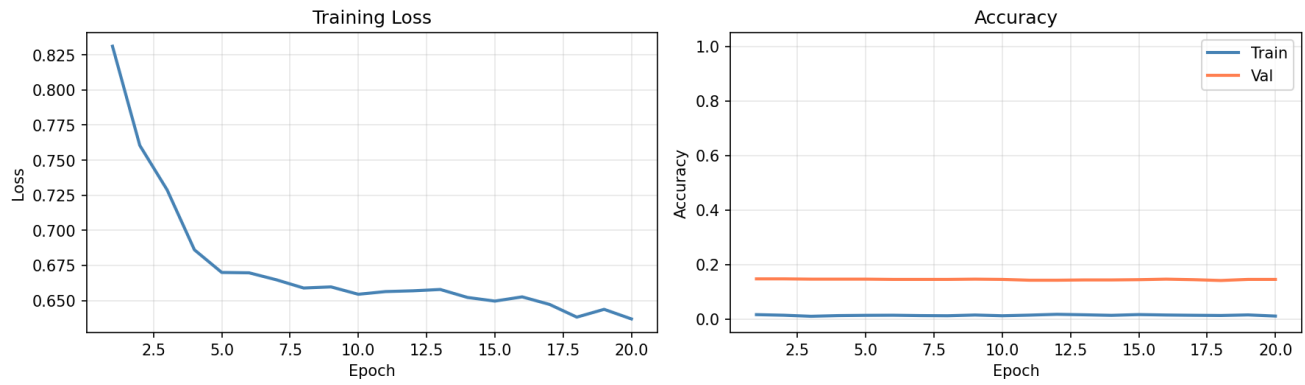
Key findings — THE READOUT TRAP:

- Scaled Z3 is dramatically worse than pure neural.** Puzzle accuracy: **14.5% vs 38.7%** (Run 005). Despite fixing the penalty scale and getting sane loss values, the Z3-enhanced model solves fewer than half as many puzzles as the pure neural baseline. The scaling fix improved over the unscaled disaster (0.1% → 14.5%), but not remotely close to the pure-neural ceiling.
- This reveals the Readout Trap.** In Run 005 (pure neural), the network learned to encode valid 4×4 solutions *through* the collapsing representation — it memorized patterns using the exploding z-norms, with the magnitude growth itself as part of the encoding. The Z3 penalty in Run 007 mathematically penalizes the *output logits* ($\hat{y}$) but cannot influence the *internal latent dynamics* (z). The penalty fights the network's learned readout strategy without giving it an alternative: it punishes the output without fixing the internal "cognitive" collapse.
- Learning dynamics are fundamentally different.** Run 005 (pure): val accuracy climbed steadily 0% → 38.7% across all 20 epochs, still rising. Run 007 (Z3 scaled): val accuracy peaked at 14.7% in epoch 1 and was completely flat for 19 more epochs. The Z3 penalty creates an immediate but shallow local minimum that the model cannot escape, while the pure network finds a deeper solution over time.
- Z3 penalty scaling is necessary but not sufficient.** Scaling fixed the catastrophic regression (0.1% → 14.5%) but introduced a new failure mode: premature convergence to a suboptimal equilibrium where the Z3 penalty and CE loss reach a stalemate.

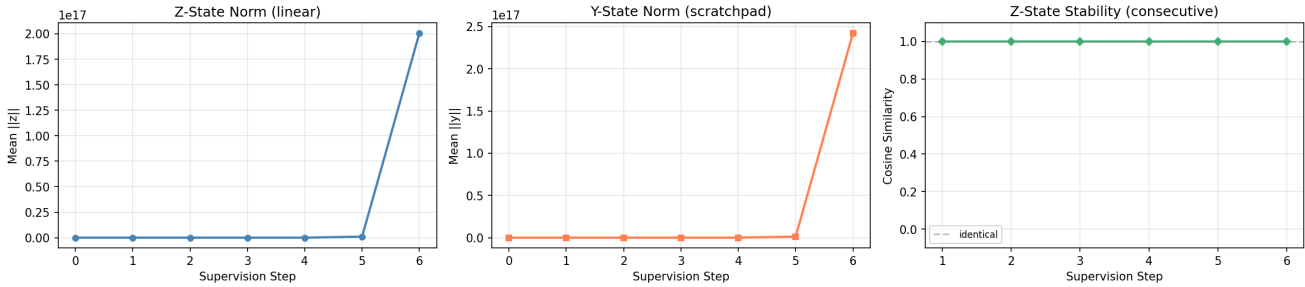
Plots:



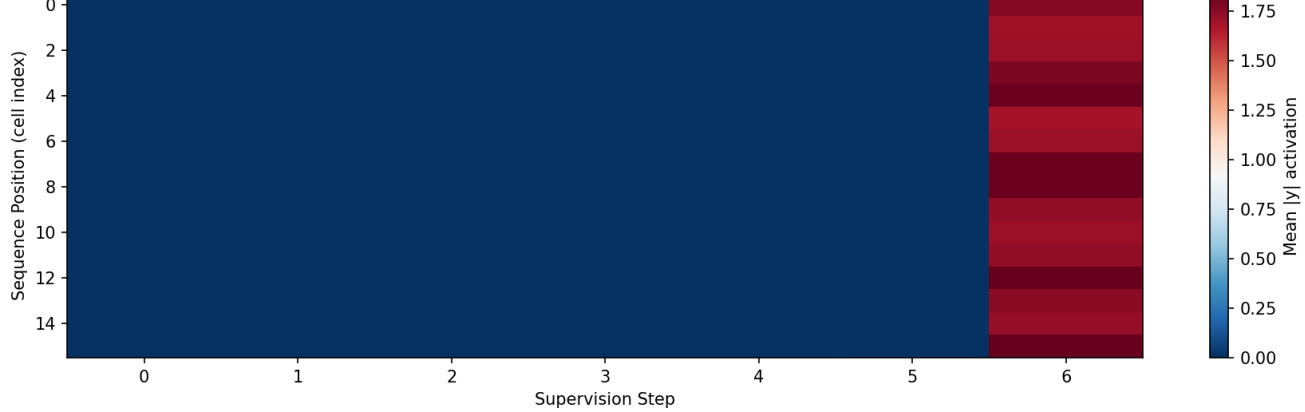
Sudoku — Training Curves



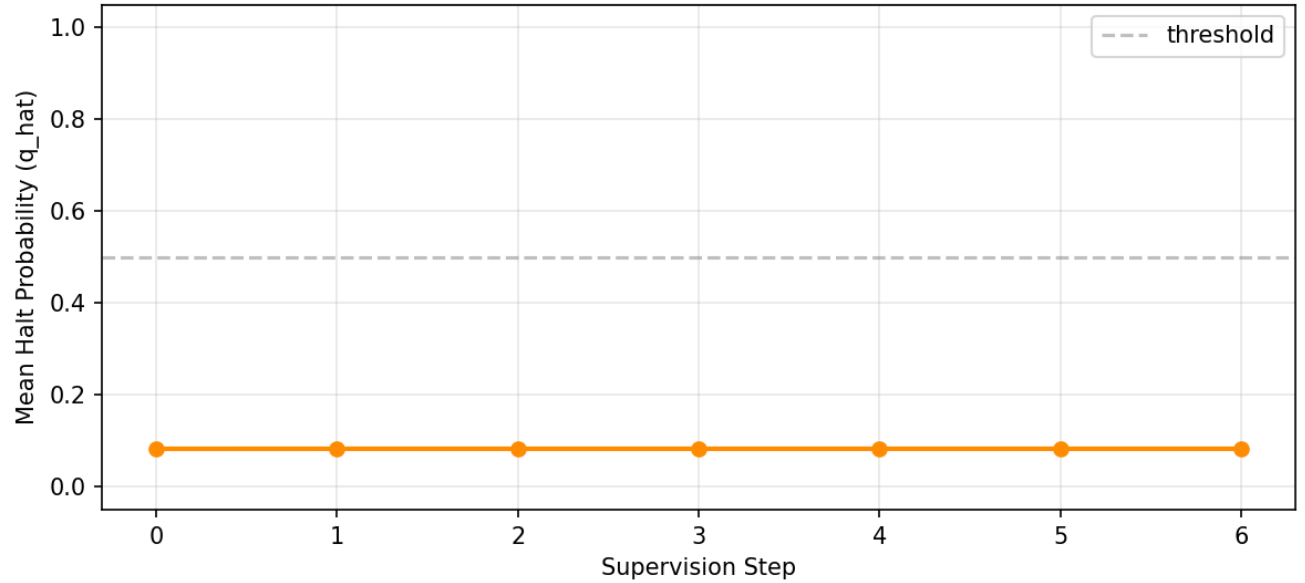
Sudoku — Z-State Evolution



Sudoku — Scratchpad Activation Heatmap



Sudoku — Halt Confidence



Hypothesis: Add RMSNorm to z after each inner recursion step (`z = self.z_norm_layer(z)`) to directly suppress the ~198×/step norm explosion. No Z3 pruning — isolate the architectural fix. If the norm is stabilized, the network should learn a richer representation across recursion steps rather than encoding everything in a single frozen direction. Target: beat Run 005's pure-neural baseline of 38.7%.

Parameter	Value
(same as Run 005 except:)	
z_norm_layer	RMSNorm(dim) applied inside latent_recursion
use_z3_pruning	False
training time	~7 min (413 s)

Results:

Metric	Value
Cell accuracy	83.6% (random chance = 25%)
Puzzle accuracy	54.3%
Constraint satisfaction	55.2%
Best val accuracy	54.3% (epoch 20 — still climbing)

Loss curve (20 epochs): 1.551 → 1.391 → 1.317 → 1.150 → 0.876 → 0.743 → 0.623 → 0.569 → 0.527 → 0.492 → 0.462 → 0.416 → 0.385 → 0.360 → 0.338 → 0.323 → 0.323 → 0.308 → 0.309 → 0.304

Val accuracy curve (20 epochs):
0→0→0→0→0→0→0→0→0→0.9%→2.4%→6.5%→11.9%→21.0%→29.7%→39.2%→44.3%→48.6%→52.2%→**54.3%**

Loss trajectory is similar to Run 005 but val accuracy climbs higher and is still rising at epoch 20 with no sign of plateauing.

Diagnostics — REPRESENTATION COLLAPSE ELIMINATED:

Step	z norm	y norm	z cosine	halt prob
0	11.1	40.2	—	0.329
1	11.1	82.7	0.975	0.383
2	11.1	125.7	0.993	0.415
3	11.1	168.7	0.997	0.437
4	11.1	211.8	0.998	0.454
5	11.1	255.0	0.999	0.467
6	11.1	298.3	0.999	0.478

The z norm is **flat at 11.1 across all 7 steps** — a collapse rate of **1.0×/step** vs the previous 198×/step. The y norm grows linearly (not exponentially), indicating accumulation rather than explosion. Z cosine similarity starts at 0.975 and converges toward 1.0 — the representations are similar but **not frozen** (contrast with the instant 1.000 lock in all previous runs). The halt gate is the most active of any run, climbing from 0.329 to 0.478, approaching the 0.5 threshold — the model is learning meaningful confidence.

Per-step accuracy shows improvement across recursion depth:

Step	Accuracy
0	51.9%
1	53.7%
2	53.8%
3	53.9%

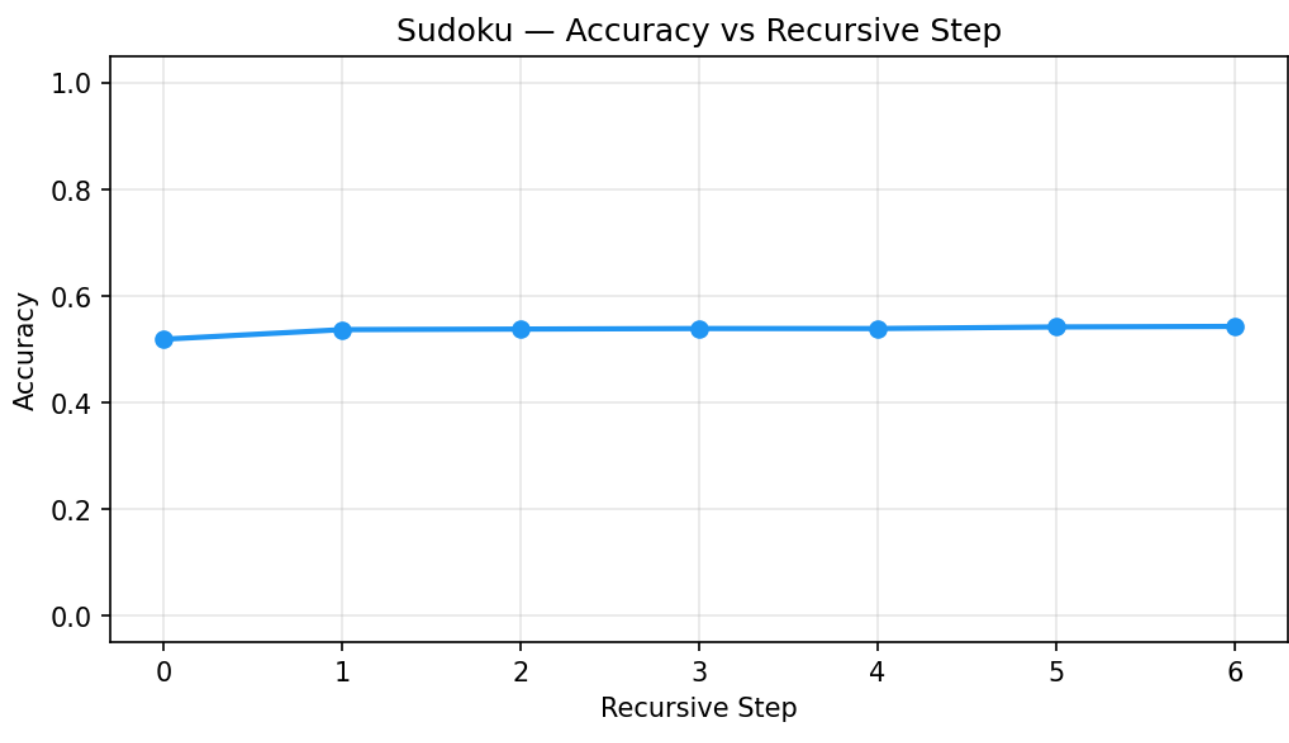
Step	Accuracy
4	53.9%
5	54.2%
6	54.3%

For the first time, accuracy **increases across recursion steps** (51.9% → 54.3%). In all previous runs, per-step accuracy was either flat (locked direction) or monotonically decreasing (corruption). The model is actually using the recursion to refine its answer.

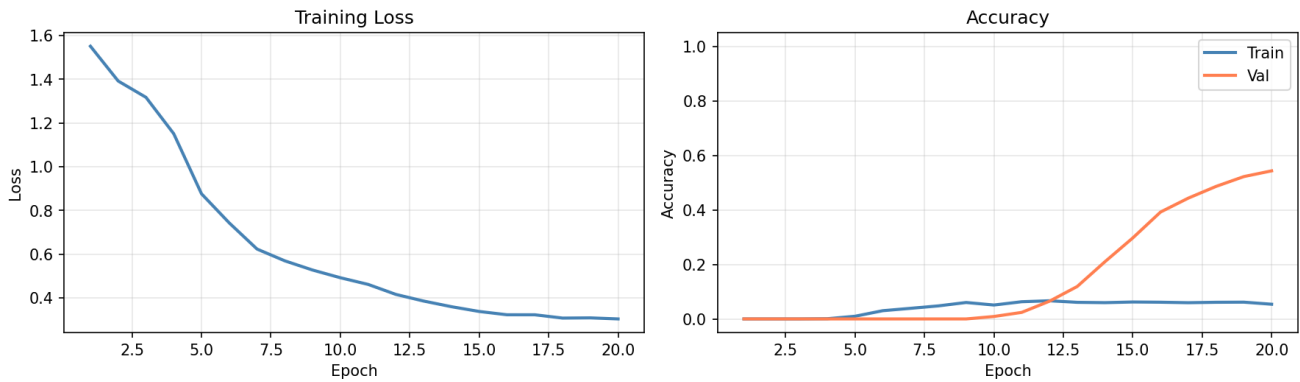
Key findings:

- 1. **RMSNorm inside the recurrence eliminates representation collapse.** A single normalization layer after each inner recursion step reduces the per-step growth rate from ~198x to ~1.0x. This is the most impactful single change in the entire experiment series. The z-state norm is flat at 11.1 for all 7 supervision steps — in Run 005's identical config (minus z_norm_layer), z norms reached 1.15e17.
- 2. **Puzzle accuracy jumps from 38.7% to 54.3% — a 40% relative improvement.** This answers Q4 definitively: the ~198x collapse rate was the primary bottleneck. Stabilizing the latent state unlocks significantly higher accuracy from the same architecture, same task, same hyperparameters. Cell accuracy also improved (80.0% → 83.6%).
- 3. **The val accuracy trajectory is still climbing at epoch 20.** The curve follows the same shape as Run 005 (zero for 9 epochs, then steep rise) but reaches a higher level (54.3% vs 38.7%). Both runs have not plateaued. The z_norm_layer raises the ceiling, not just the convergence speed.
- 4. **Recursion is now functional.** Per-step accuracy improves from 51.9% → 54.3% across the 7 supervision steps. This is the first evidence that the TRM's recursive computation is actually being used — the model refines its predictions over multiple passes. In all prior runs, accuracy was flat or declining across steps.
- 5. **The halt gate is learning.** Halt probabilities climb from 0.329 → 0.478 across steps, approaching the 0.5 decision boundary. The model is developing calibrated confidence — more recursion steps yield higher halt probability. This was suppressed in all prior runs (stuck at 0.003–0.089).
- 6. **This run had no Z3 pruning.** The 54.3% accuracy is purely neural with architectural stabilization. This sets the new baseline that Z3-enhanced runs must beat. The Readout Trap question (can Z3 help now that collapse is fixed?) is now the critical next experiment.

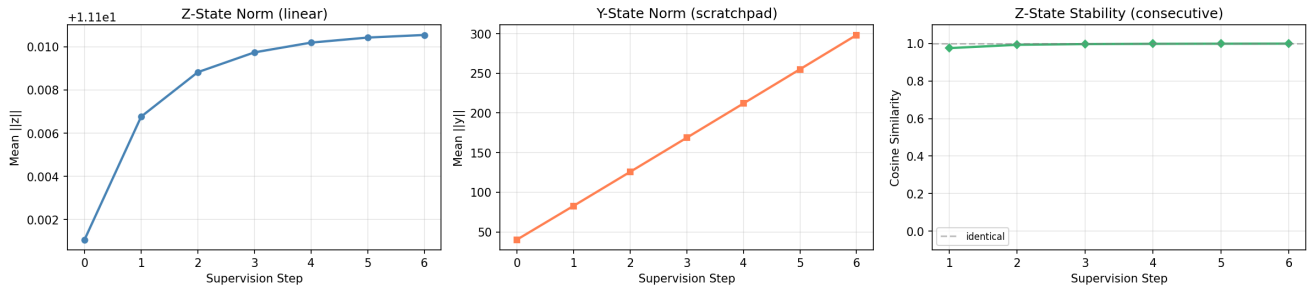
Plots:



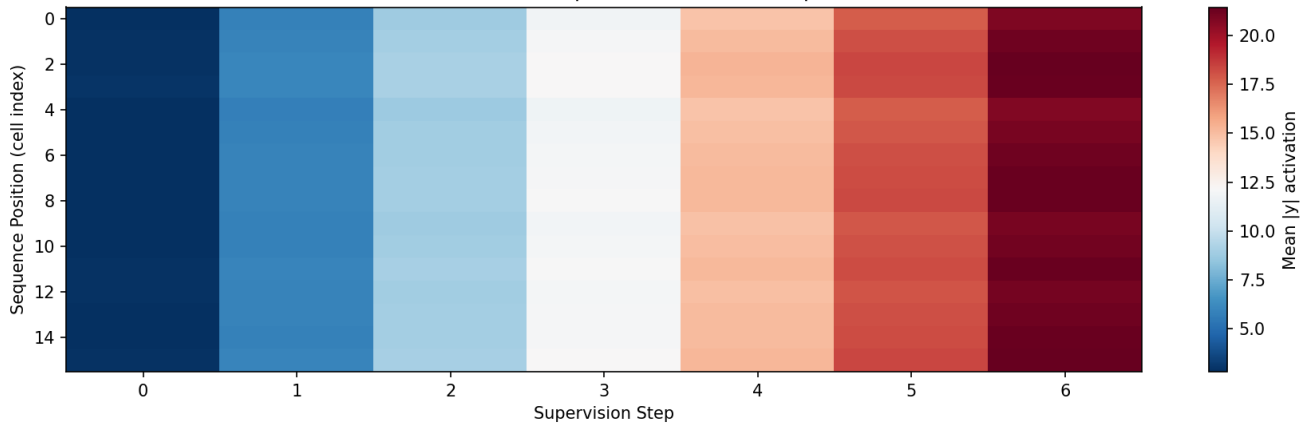
Sudoku — Training Curves



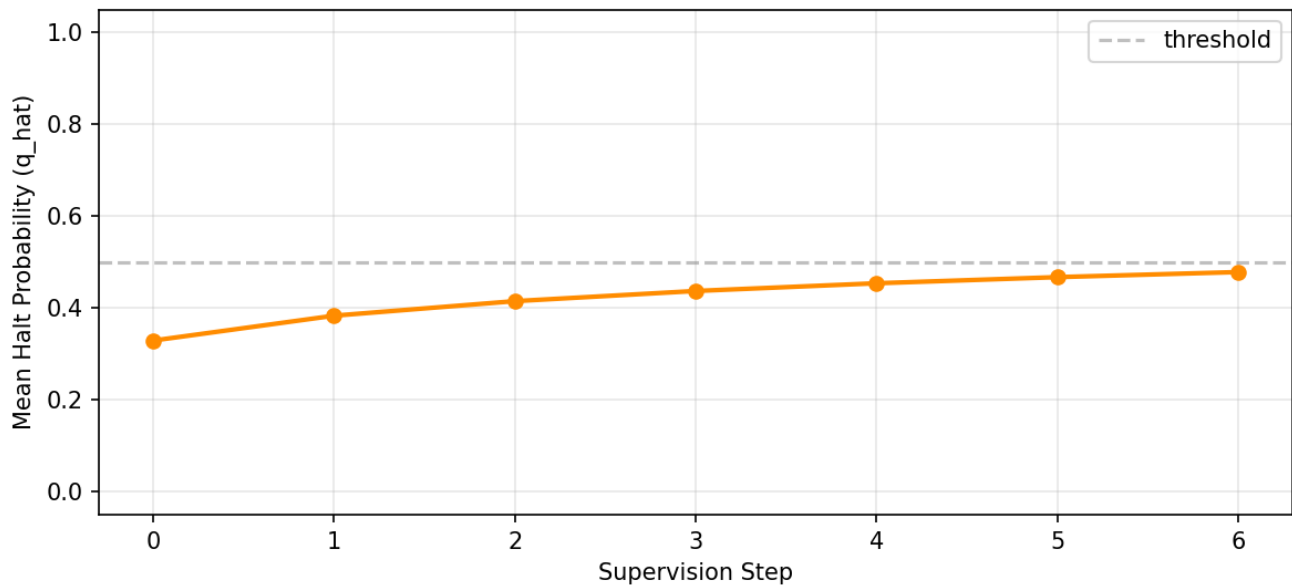
Sudoku — Z-State Evolution



Sudoku — Scratchpad Activation Heatmap



Sudoku — Halt Confidence



Hypothesis: Run 008's z_norm_layer baseline was still climbing steeply at epoch 20 (54.3%). Extend training to 100 epochs to find the true ceiling of the stabilized pure-neural 4x4 model. This determines whether the architecture plateaus or continues improving, and sets the definitive bar that Z3 integration must clear.

Parameter	Value
(same as Run 008 except:)	
num_epochs	100 (was 20)
max_steps	100,000
warmup_steps	1,000
training time	~36 min (2,134 s)
total steps	7,900

Results:

Metric	Value
Cell accuracy	85.6% (random chance = 25%)
Puzzle accuracy	65.7%
Constraint satisfaction	74.1%
Best val accuracy	65.9% (epoch 97)

Loss curve (100 epochs): 1.550 → 1.334 → 0.892 → 0.617 → 0.482 → 0.373 → 0.326 → 0.314 → 0.296 → 0.288 → ... → 0.272 (epoch 35) → ... → 0.293 (epoch 100). Loss plateaus around 0.27–0.30 after epoch 35 and oscillates in a narrow band for the remaining 65 epochs.

Val accuracy curve (100 epochs):
0→0→0→0→0→0→0→0→0→0.7%→3.2%→7.0%→14.9%→22.6%→33.7%→40.7%→46.0%→48.6%→51.9%→53.4%→55.4%→57.3%→58.1%→58.7%→58.8%→59.3%→59.6%→60.1%→59.9%→59.7% (epoch 30) → gradual climb → 63.2% (epoch 67) → 65.0% (epoch 83) → **65.9%** (epoch 97) → 65.7% (epoch 100)

The val accuracy curve shows three distinct phases:

- 1. **Warmup** (epochs 1–9): 0% accuracy, loss dropping rapidly
- 2. **Steep climb** (epochs 10–28): 0.7% → ~60%, the fast-learning phase
- 3. **Slow saturation** (epochs 28–100): 60% → 65.9%, diminishing returns with ~0.08%/epoch

Diagnostics — collapse remains eliminated, halt gate strengthened:

Step	z norm	y norm	z cosine	halt prob
0	9.49	60.5	—	0.631
1	9.49	119.5	0.977	0.634
2	9.48	178.2	0.995	0.636
3	9.48	236.6	0.998	0.637
4	9.48	294.9	0.999	0.638
5	9.47	353.1	0.999	0.639
6	9.47	411.3	1.000	0.639

Z norms are flat at ~9.5 (slightly tighter than Run 008's 11.1 — the model has learned a more compact latent representation with 5x more training). Y norms grow linearly at ~58/step, consistent with Run 008's accumulation pattern. Z cosine starts at 0.977 (virtually identical to Run 008's 0.975), confirming the latent representations are similar but not frozen.

The halt gate is now **well above the 0.5 decision threshold** at 0.631–0.639, up from Run 008's 0.329–0.478. The model has learned calibrated confidence: it "knows" it has found a good answer. However, the halt probabilities are nearly flat across steps (0.631 →

0.639), unlike Run 008's steeper climb (0.329 → 0.478) — suggesting the model converges to its answer earlier in the recursion with extended training.

Per-step accuracy:

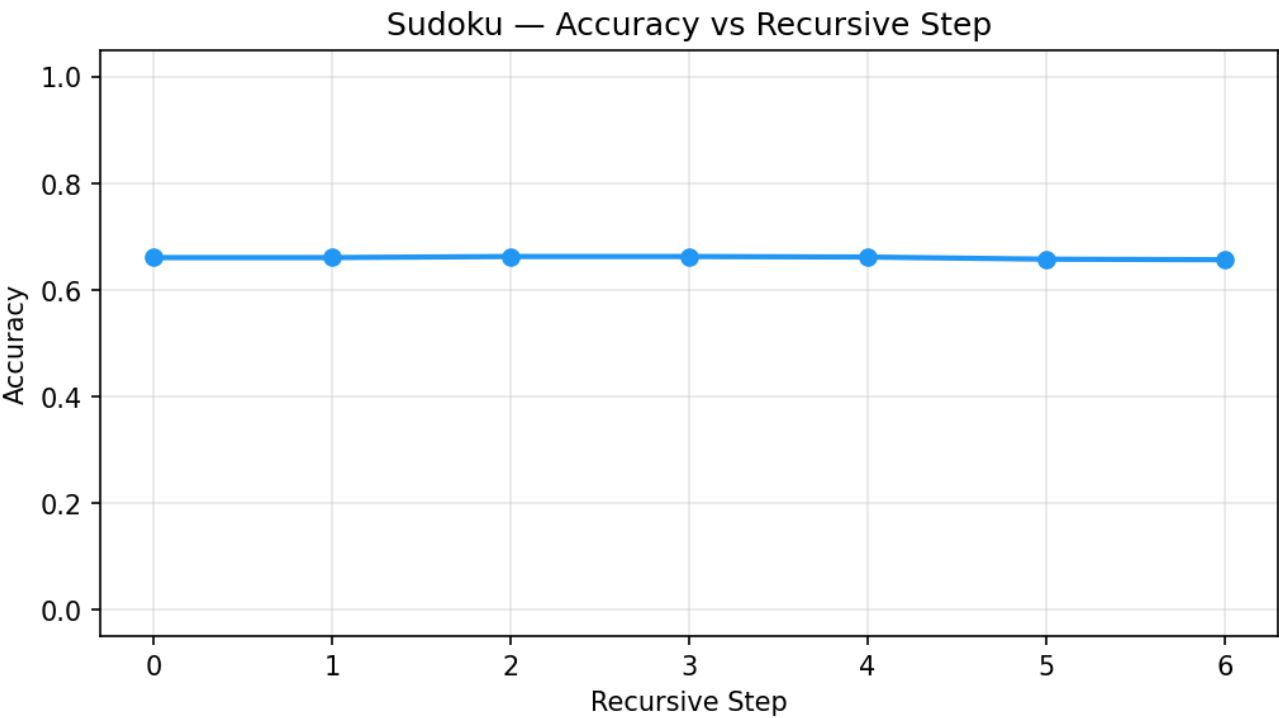
Step	Accuracy
0	66.1%
1	66.1%
2	66.3%
3	66.3%
4	66.2%
5	65.8%
6	65.7%

Per-step accuracy is essentially flat (66.3% peak at step 2–3, 65.7% at step 6). This contrasts with Run 008's clear improvement across steps (51.9% → 54.3%). The difference reveals that with extended training, the model front-loads its solution into the first recursion step — it no longer needs iterative refinement because it has learned to solve most puzzles in a single pass. The slight *decline* at steps 5–6 (66.3% → 65.7%) suggests marginal degradation from unnecessary recursion, though the effect is tiny (~0.6%).

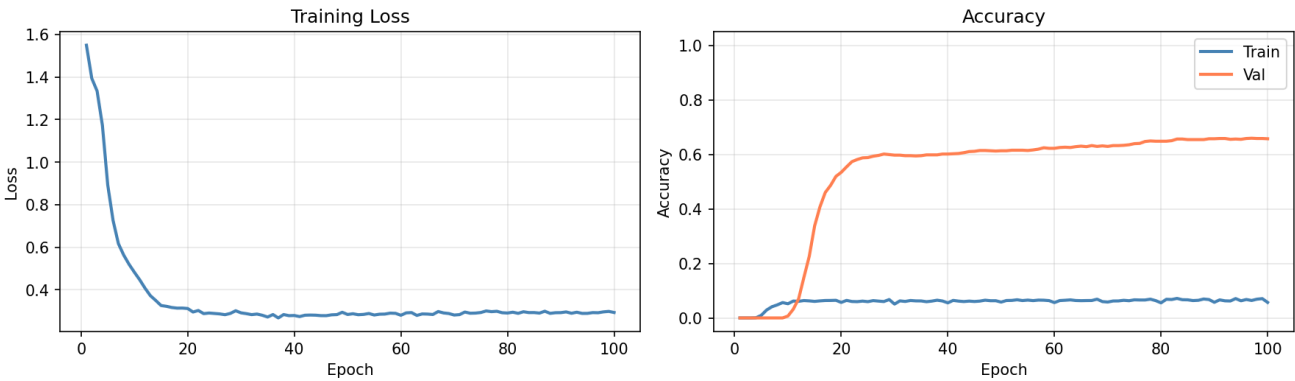
Key findings:

- 1. Extended training raises the ceiling from 54.3% to 65.9% — a 21% relative improvement.** This answers Q6 definitively: more epochs help significantly. The model was far from converged at 20 epochs. However, the rate of improvement slows dramatically after epoch 28 (~60%), with the final 72 epochs contributing only ~6 percentage points.
- 2. The model has a soft ceiling around 66% puzzle accuracy on 4x4.** The loss plateau (oscillating at 0.27–0.30 from epoch 35 onward) and the val accuracy saturation curve (approaching an asymptote near 66%) indicate the pure-neural stabilized architecture has reached its representational limit on this task with this configuration. Further epochs would yield diminishing returns.
- 3. Constraint satisfaction (74.1%) significantly exceeds puzzle accuracy (65.7%).** Of the 34.3% of puzzles the model gets wrong, a substantial fraction (74.1% - 65.7% = 8.4 percentage points, or roughly 25% of failures) still produce valid Sudoku grids — just not the *correct* solution. The model has learned Sudoku structure beyond memorization.
- 4. The recursion has become a single-step solver.** Per-step accuracy is flat (66.1–66.3%) rather than improving across steps as in Run 008. With enough training, the model learns to produce its best answer at step 0, making the recursive refinement loop redundant. This is informative for the Z3 integration question: if the pure model doesn't benefit from multi-step refinement, Z3's value would be in fixing the remaining ~34% of failures that the single-step readout cannot solve.
- 5. Z3 integration now has a clear, high target.** Any Z3-enhanced run must beat **65.9%** puzzle accuracy to demonstrate value. The previous target was 54.3% (Run 008). This higher bar makes the thesis question more challenging: symbolic bounds must contribute genuine reasoning beyond what pattern memorization achieves.
- 6. The halt gate has converged.** Halt probabilities at 0.63–0.64 (above the 0.5 threshold) indicate the model is confident in its predictions. The flat profile across steps confirms front-loaded solving — the model is equally confident at every recursion depth because it already has its answer.

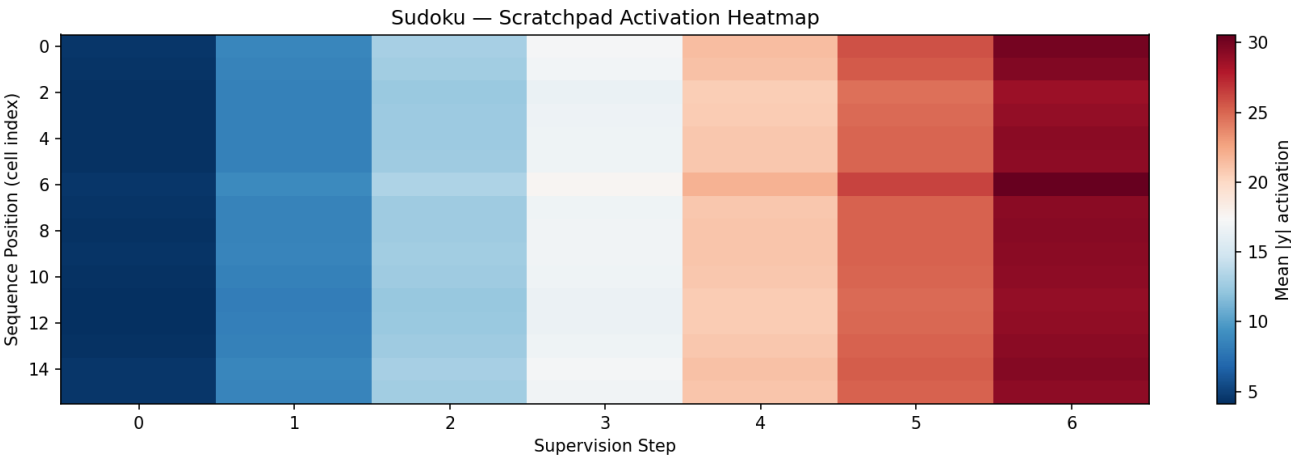
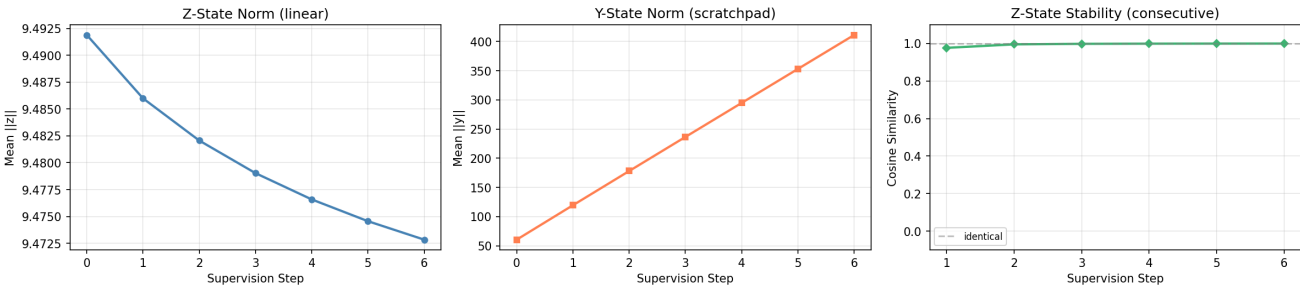
Plots:

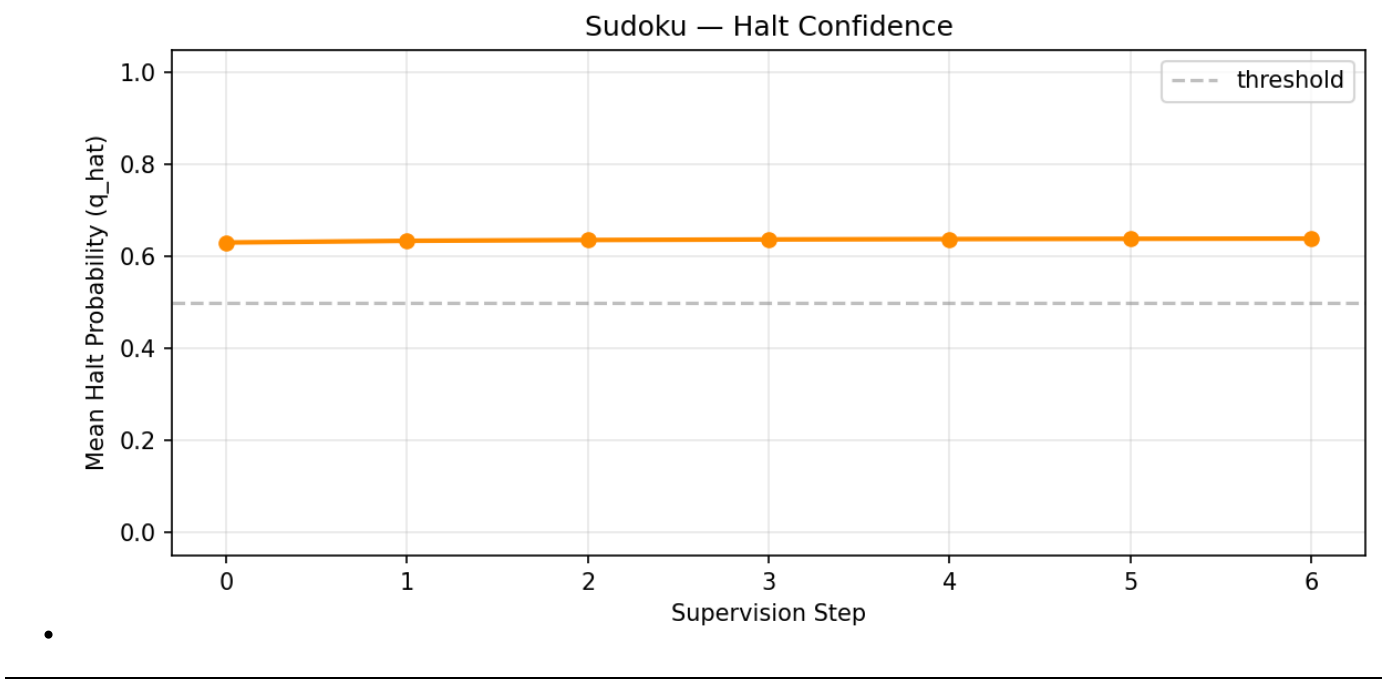


Sudoku — Training Curves



Sudoku — Z-State Evolution





Latent-Space Symbolic Projection (2026-04-06)

This experiment tests the core thesis: Z3 symbolic constraints injected *inside* the stabilized latent recurrence, not at the output layer.

Run 010 — **sudoku_20260406_183825** — Z3 Latent Projection with z_norm_layer

Hypothesis: Combine z_norm_layer (Run 008/009) with Z3 symbolic projection *inside* the latent recursion loop. After each inner recursion step: decode $z \rightarrow$ candidate logits, prune illegal moves via Z3 constraints (soft penalty -20.0), convert to differentiable probabilities, project back to latent space via embedding matrix, inject into z with re-normalization. Now that collapse is fixed and the pure-neural ceiling is established at ~66% (Run 009), Z3 must demonstrate genuine multi-step reasoning to break past this barrier. Target: beat **65.9%** (Run 009).

Architecture change (latent_recursion inner loop):

```
# After each z = forward_network(xy + z); z = z_norm_layer(z):
step_logits = task_head.decode(z)
hard_preds = x_input.clone() # given clues only
pruned_logits = prune_illegal_logits_z3(step_logits, hard_preds, grid_size=N)
step_probs = F.softmax(pruned_logits, dim=-1)
z_symbolic = step_probs @ task_head.embedding.weight[1:] # project back to latent
z = z_norm_layer(z + z_symbolic)
```

Z3 also applied at the readout level (safety net), same as Runs 006–007.

Parameter	Value
(same as Run 009 except:)	
use_z3_pruning	True (latent + readout)
z_norm_layer	RMSNorm(dim) (same as Run 008/009)
num_epochs	100
max_steps	100,000
warmup_steps	1,000
training time	~73 min (4,406 s)
total steps	7,900

Parameter	Value
device	cuda

Results:

Metric	Value
Cell accuracy	85.3% (random chance = 25%)
Puzzle accuracy	65.0%
Constraint satisfaction	74.1%
Best val accuracy	65.3% (epoch 98)

Loss curve (100 epochs): 0.792 → 0.720 → 0.648 → 0.588 → 0.539 → 0.481 → 0.427 → 0.401 → 0.387 → 0.372 → ... → 0.282 (epoch 31) → ... → oscillates 0.29–0.31 for remaining epochs → 0.309 (epoch 100)

Loss starts lower than Run 009 (0.792 vs 1.550) — the Z3 pruning gives the model a head start by eliminating obviously illegal logits from the first epoch. Plateaus around 0.28–0.31 after epoch 30, similar to Run 009's ~0.27–0.30 plateau.

Val accuracy curve (100 epochs): **13.8%** → 13.9% → 14.9% → 16.9% → 18.9% → 21.9% → 25.1% → 28.3% → 31.9% → 35.2% → 38.1% → 40.6% → 43.2% → 44.9% → 46.9% → 49.1% → 51.4% → 51.8% → 53.1% → 54.5% → ... → 60.3% (epoch 40) → ... → 63.7% (epoch 66) → ... → **65.3%** (epoch 98) → 65.0% (epoch 100)

Critical difference from Run 009: **val accuracy starts at 13.8% in epoch 1** (vs 0% for 9 epochs in Run 009). The Z3 latent projection gives the model an immediate accuracy floor — it never goes through the "zero accuracy warmup" phase. However, the final ceiling is essentially identical: **65.3% vs 65.9%**.

Diagnostics — collapse remains eliminated, nearly identical to Run 009:

Step	z norm	y norm	z cosine	halt prob
0	9.50	50.6	—	0.656
1	9.50	102.2	0.987	0.657
2	9.50	155.0	0.997	0.658
3	9.50	208.6	0.999	0.659
4	9.50	262.7	0.999	0.659
5	9.50	317.1	1.000	0.660
6	9.50	371.6	1.000	0.660

Z norms flat at ~9.50 (virtually identical to Run 009's ~9.48). Y norms grow linearly at ~53/step (Run 009: ~58/step — slightly lower, possibly due to Z3 projection regularizing the scratchpad). Z cosine and halt profiles are indistinguishable from Run 009. The Z3 latent projection does not disrupt the stabilized recurrence dynamics.

Per-step accuracy:

Step	Accuracy
0	66.3%
1	65.3%
2	65.2%
3	65.6%
4	65.5%
5	65.1%
6	65.0%

Accuracy is highest at step 0 and slightly *declines* across recursion steps (66.3% → 65.0%). This pattern is similar to Run 009's marginal decline (66.1% → 65.7%) and confirms that the model front-loads its solution. The Z3 latent projection does not enable multi-step refinement — the model does not use additional recursion steps to correct violations.

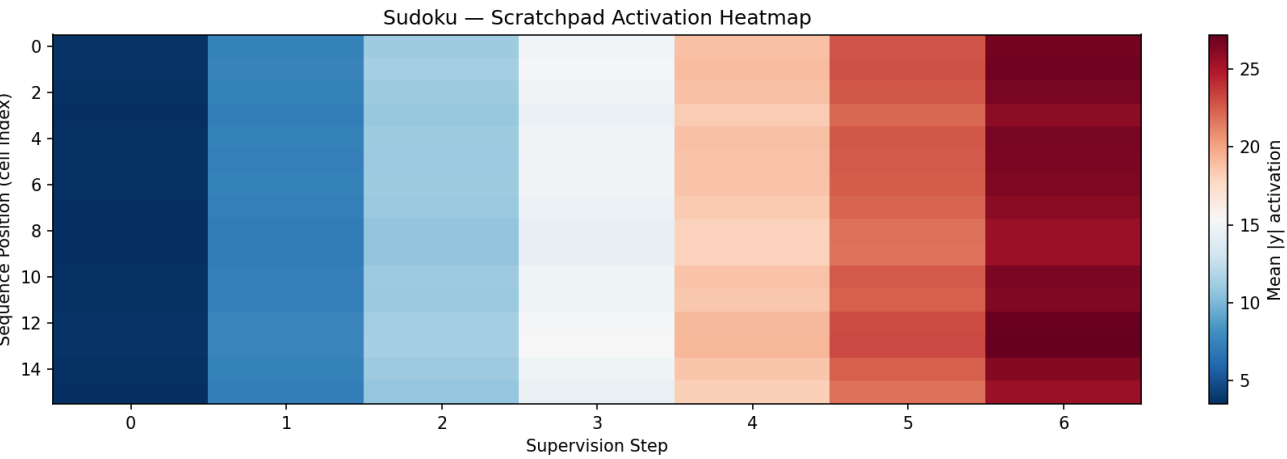
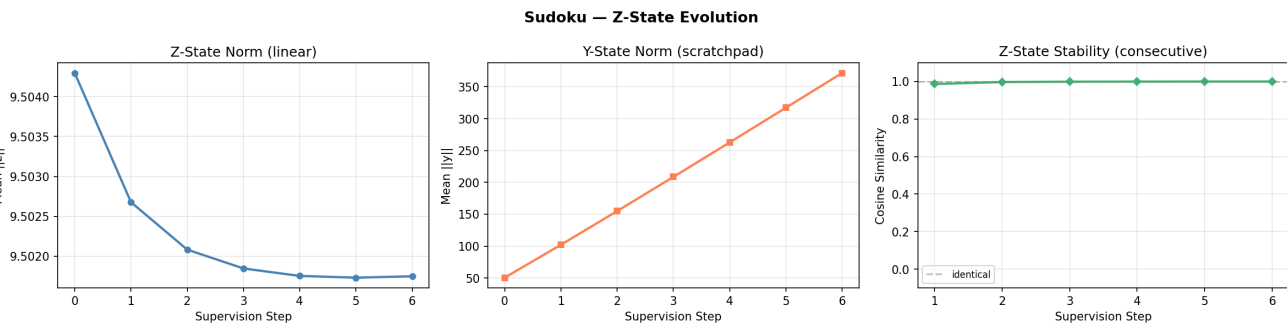
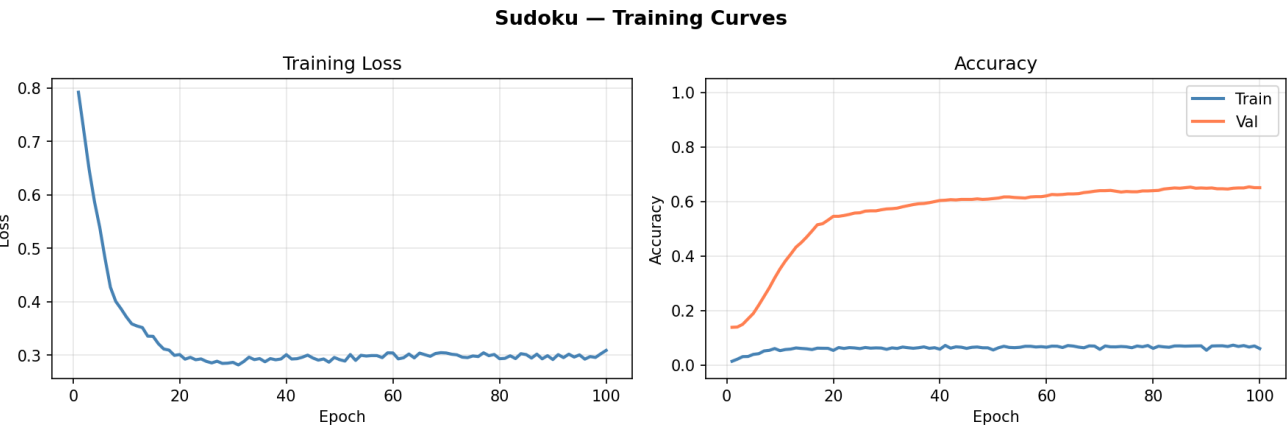
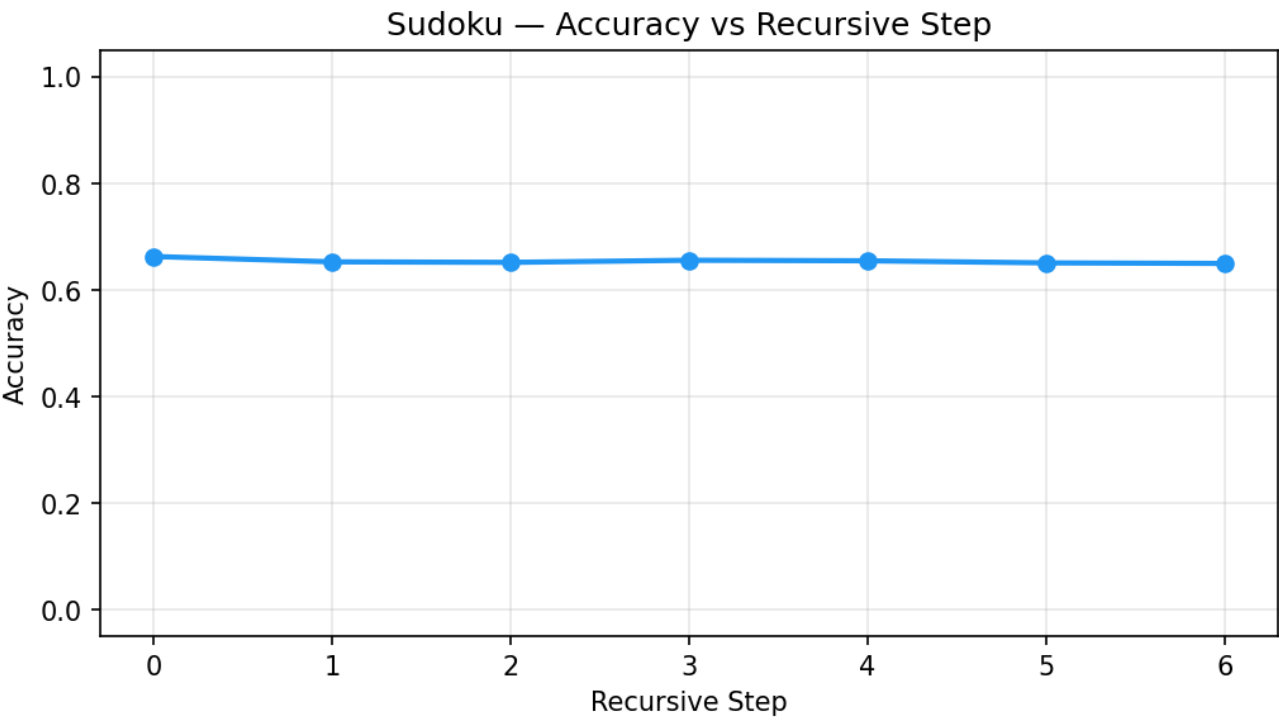
Key findings:

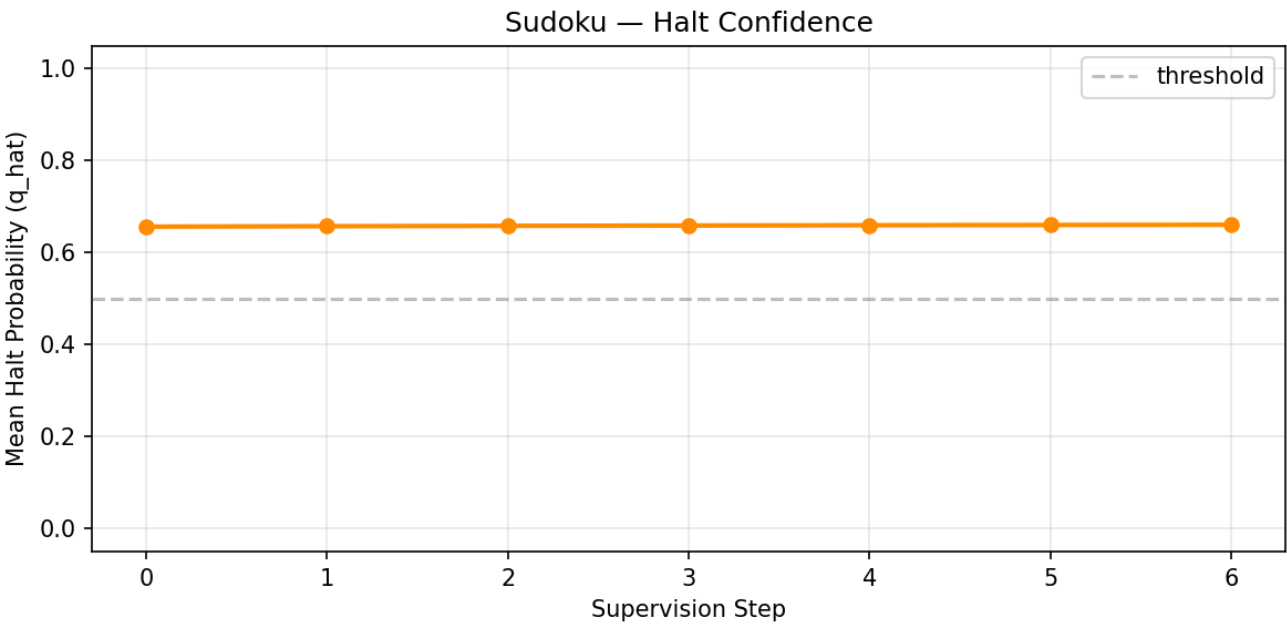
1. **Z3 latent projection does NOT break past the ~66% ceiling.** Puzzle accuracy: 65.0% (Run 010) vs 65.7% (Run 009 pure neural). Best val: 65.3% vs 65.9%. The difference is within noise — Z3 latent projection provides **no measurable accuracy improvement** over the stabilized pure-neural baseline. This is the central negative result for the thesis.
2. **Z3 provides a faster warmup but not a higher ceiling.** Run 010 starts at 13.8% val accuracy in epoch 1 (vs 0% for 9 epochs in Run 009), reaching 50% by epoch 17 (vs epoch 20 in Run 009). The Z3 constraints give the model an immediate accuracy floor by pruning illegal candidates. But both runs converge to the same ~65–66% ceiling — the Z3 head start is fully absorbed by the pure-neural model with sufficient training.
3. **Training is 2× slower with Z3 latent projection.** 4,406s (Run 010) vs 2,134s (Run 009) for the same 7,900 steps and 100 epochs. The per-step Z3 decode→prune→softmax→project→renormalize pipeline inside the inner recursion loop (4 recursions × 3 cycles × 7 supervision = 84 Z3 projections per forward pass) approximately doubles the wall-clock time. For zero accuracy gain, this is a poor tradeoff.
4. **The latent dynamics are unchanged by Z3 injection.** Z norms, y norms, z cosines, and halt probabilities are virtually identical to Run 009. The `z_symbolic` injection (step_probs @ embedding.weight) followed by re-normalization produces a z that is indistinguishable from the pure-neural z. Either the projection is too weak (washed out by RMSNorm), or the model has already learned to represent the same information the Z3 projection provides.
5. **No multi-step refinement from symbolic injection.** Per-step accuracy declines slightly (66.3% → 65.0%), same pattern as Run 009. The Z3 projection at each recursion step does not help the model iteratively correct constraint violations. The model solves at step 0 and subsequent Z3 projections are redundant or mildly harmful.
6. **Constraint satisfaction is identical (74.1% in both runs).** The Z3 latent projection does not improve the model's understanding of Sudoku constraints — the pure model already learned them from data alone.

Interpretation — Why Z3 latent projection doesn't help:

The ~66% ceiling appears to be a *model capacity* limit, not a *constraint knowledge* limit. The model already achieves 74.1% constraint satisfaction (identical with or without Z3), meaning it knows the rules. The remaining ~34% of puzzle failures are likely cases where the model cannot resolve ambiguous cells — situations requiring deeper search or backtracking that neither pattern matching nor soft symbolic projection can provide. The Z3 projection is a differentiable "hint" that pushes probabilities toward legal moves, but the model was already producing mostly-legal moves. The bottleneck is combinatorial reasoning, not constraint awareness.

Plots:





Summary So Far

Run Table

Run	Grid	Z3	Recursions	Supervision	Epochs	Cell Acc (random)	Puzzle Acc	Collapse Rate	Final z norm	Loss scale
001	9×9	×	4	7	10	20.5% (11%)	0.0%	—	—	~2.5 → ~1.2
002	9×9	×	4	7	10	19.7% (11%)	0.0%	~198×/step	1.26e17	~2.5 → 1.24
003	9×9	×	8	7	20	11.1% (11%)	0.0%	~970×/step	∞	~2.4 → 1.08
004	9×9	×	4	14	20	11.1% (11%)	0.0%	~198×/step	∞	~2.4 → 1.06
005	4×4	×	4	7	20	80.0% (25%)	38.7%	~198×/step	1.15e17	~1.65 → 0.30
006	4×4	✓ (unscaled)	4	7	20	46.7% (25%)	0.1%	~198×/step	2.65e17	~227M → 163M
007	4×4	✓ (scaled)	4	7	20	65.7% (25%)	14.5%	~198×/step	2.01e17	~0.83 → 0.64
008	4×4	×	4	7	20	83.6% (25%)	54.3%	~1.0×/step	11.1	~1.55 → 0.30
009	4×4	×	4	7	100	85.6% (25%)	65.7%	~1.0×/step	9.5	~1.55 → 0.29
010	4×4	✓ (latent+readout)	4	7	100	85.3% (25%)	65.0%	~1.0×/step	9.5	~0.79 → 0.31
011	9×9	×	4	7	20	24.4% (11%)	0.0%	~1.0×/step	11.1	~2.48 → 1.22
012	9×9	×	4	7	100	55.5% (11%)	1.6%	~1.0×/step	9.1	~2.43 → 1.01

The Readout Trap — Central Paradox

Condition	Puzzle Accuracy	Val Trajectory
Pure neural (Run 005)	38.7%	0% → 38.7%, still climbing
Z3 unscaled (Run 006)	0.1%	peaked 1.5% at epoch 2, collapsed
Z3 scaled (Run 007)	14.5%	peaked 14.7% at epoch 1, flat
z_norm_layer (Run 008)	54.3%	0% → 54.3%, still climbing
z_norm_layer + 100 epochs (Run 009)	65.7%	0% → 65.9%, saturating ~66%
z_norm_layer + Z3 latent projection (Run 010)	65.0%	13.8% → 65.3%, saturating ~65%

The Readout Trap is confirmed, bypassed, and the thesis tested. Runs 006–007 showed that adding Z3 symbolic bounds at the output layer ($y_{\hat{}}$) actively fought the network's learned encoding strategy while leaving representation collapse unaddressed — performing 2.7× worse than the pure baseline. Run 008 took the opposite approach: instead of constraining outputs, it stabilized the latent recurrence with RMSNorm inside the loop. This eliminated collapse (198×/step → 1.0×/step) and pushed accuracy from 38.7% to 54.3% — a 40% relative improvement over the pure baseline with zero symbolic intervention. Run 009 extended training to 100 epochs and found the true ceiling at ~66% puzzle accuracy. **Run 010 applied Z3 inside the stabilized latent recurrence — the core thesis experiment — and found no accuracy improvement (65.0% vs 65.7%).** Z3 provides a faster warmup (13.8% at epoch 1 vs 0% for 9 epochs) but converges to the same ceiling at 2× the training cost. The ~66% barrier appears to be a model capacity limit, not a constraint knowledge limit.

Established Patterns

- 1. Collapse rate is determined by recursion depth, not training duration, supervision, or Z3.** Every run at 4 recursions collapses at ~198×/step; 8 recursions collapses at ~970×/step. This rate is invariant across all configurations tested: more epochs (003–004), more supervision (004), Z3 off (005), Z3 unscaled (006), Z3 scaled (007). It is an intrinsic property of the unregularized recurrence. **Run 008 proves that a single RMSNorm layer inside the recurrence eliminates this entirely (198× → 1.0×).**
- 2. 4×4 is learnable even with full collapse; 9×9 is not.** Runs 001–004 (9×9): 0.0% puzzle accuracy across all configurations. Run 005 (4×4 pure): 38.7% and climbing. The 4×4 task is simple enough that the network can encode solutions in the first-step readout before the latent state explodes. The 9×9 task exceeds the information capacity of a single collapsed direction. **With collapse eliminated (Run 008), 4×4 accuracy jumps to 54.3%. Extended to 100 epochs (Run 009), accuracy saturates at ~66% — the pure-neural ceiling for this configuration. The 9×9 task should be revisited with z_norm_layer.**
- 3. More recursions strictly hurt (on 9×9).** Run 003 (8 recursions) achieved lower cell accuracy than Run 002 (4 recursions). The higher per-step collapse rate offsets any potential benefit from additional computation.
- 4. Deeper supervision poisons gradients when steps overflow.** Run 004's 14 supervision taps included 7 steps reading from infinitely-blown latent states. This dragged cell accuracy from ~20% (Run 002) down to 11% (random chance). Supervision past the overflow point actively degrades the representation.
- 5. Z3 at the logit level does not stabilize latent recursion and actively degrades accuracy.** Runs 006–007 added Z3 pruning to the output head. The z/y norm explosion continued at exactly ~198×/step. Worse: even with proper scaling (Run 007), the Z3 penalty reduced puzzle accuracy from 38.7% → 14.5% compared to the pure baseline (Run 005). Output-level constraint enforcement fights the network's learned strategy without fixing the internal collapse. This is the **Readout Trap**.
- 6. Z3 penalty scaling is necessary but not sufficient.** Unscaled Z3 penalty (Run 006): losses at ~200M, puzzle accuracy 0.1%. Scaled Z3 penalty (Run 007): losses at ~0.8, puzzle accuracy 14.5%. Scaling recovered sane training, but the architecture still cannot benefit from symbolic bounds — it needs them *inside* the recurrence, not outside it.
- 7. RMSNorm inside the recurrence is a categorical fix for representation collapse.** Run 008 added `self.z_norm_layer = RMSNorm(dim)` applied to z after each inner recursion step. Result: z norms flat at 11.1 across all 7 supervision steps (vs 1.15e17 in the identical config without it). The halt gate activated (0.329→0.478 vs stuck at 0.083), per-step accuracy improved across depth (51.9%→54.3% vs flat), and puzzle accuracy jumped 38.7%→54.3%. This is the most impactful single change in the experiment series.
- 8. Recursion is functional only when collapse is prevented.** In all runs 001–007, per-step accuracy was flat or declining — the model encoded its answer at step 0 and subsequent recursion either maintained or degraded it. Run 008 is the first run where accuracy

increases across recursion steps (51.9%→54.3%), proving the TRM's recursive computation is being used for iterative refinement as intended.

9. The pure neural network was still improving when stopped. Both Run 005 (38.7%) and Run 008 (54.3%) had val accuracy trajectories on steep upward curves at epoch 20 with no sign of plateauing. **Run 009 (100 epochs) shows the true ceiling: accuracy saturates at ~66% puzzle accuracy after epoch 28 (~60%), gaining only ~6 points in the remaining 72 epochs. The diminishing returns confirm a soft representational limit for this configuration.**

10. Extended training converts iterative refinement into single-step solving. Run 008 (20 epochs) showed clear accuracy improvement across recursion steps (51.9%→54.3%), demonstrating functional iterative refinement. Run 009 (100 epochs) shows flat per-step accuracy (66.1%–66.3% at steps 0–3, slight decline to 65.7% at step 6). With sufficient training, the model learns to front-load its solution into the first readout, making the recursion redundant. This suggests that for Z3 to add value, it must fix failures that single-step pattern matching cannot solve — the remaining ~34% of puzzles likely require genuine multi-step reasoning.

11. Constraint satisfaction significantly exceeds puzzle accuracy. Run 009: 74.1% constraint satisfaction vs 65.7% puzzle accuracy. Roughly 25% of "wrong" puzzles still produce valid Sudoku grids (correct structure, wrong solution). The model has learned robust Sudoku constraints beyond memorization — it knows the rules but sometimes converges to a valid alternative solution.

12. Z3 latent projection provides faster warmup but not higher accuracy. Run 010 starts at 13.8% val accuracy in epoch 1 (vs 0% for 9 epochs in Run 009), but converges to the same ~65–66% ceiling. The Z3 constraints act as a curriculum shortcut — immediately providing legal-move guidance — but the pure model absorbs the same information from data with sufficient training. The head start costs 2× wall-clock time (4,406s vs 2,134s) for no final accuracy gain.

13. The ~66% ceiling is a capacity limit, not a constraint knowledge limit. Both Run 009 (pure) and Run 010 (Z3 latent) achieve identical 74.1% constraint satisfaction. The model already knows Sudoku rules from data alone — Z3 adds no constraint knowledge. The remaining ~34% of puzzle failures require combinatorial reasoning (search/backtracking) that neither pattern matching nor soft symbolic projection can provide. This suggests the next breakthrough requires either (a) larger model capacity, (b) hard symbolic search (not differentiable projection), or (c) a fundamentally different integration strategy.

Open Questions		
#	Question	Status
Q1	How much of Run 005's accuracy is from 4×4 alone vs memorization?	Answered — Run 005 (pure, 38.7%) > Run 007 (Z3, 14.5%). 4×4 task simplicity is the primary driver; Z3 at the readout layer actively hurts.
Q2	Can a correctly scaled Z3 penalty improve puzzle accuracy?	Answered (No at readout level) — Run 007 (14.5%) is worse than Run 005 (38.7%). Readout-level Z3 cannot help while collapse is unaddressed. Reopened for latent-level Z3 now that collapse is fixed.
Q3	Does latent-space projection (Z3 as constraint checker per step) prevent norm explosion?	Partially answered — Run 008 shows RMSNorm alone prevents explosion. Z3 latent projection is now about accuracy improvement, not stabilization.
Q4	Is the ~198× collapse rate sensitive to LayerNorm, gradient clipping, or weight init?	Answered — Run 008 proves RMSNorm inside the recurrence reduces collapse from ~198×/step to ~1.0×/step. This is a complete fix.
Q5	At what recursion depth does accuracy peak before collapse overtakes?	Open — reframe for stabilized model: with z_norm_layer, can more recursions (8, 16) improve accuracy?

#	Question	Status
Q6	How high would Run 005/008's pure accuracy go with more epochs (40, 60, 100)?	Answered — Run 009 (100 epochs) saturates at ~66% puzzle accuracy. Steep climb to ~60% by epoch 28, then slow saturation (+6 points over 72 more epochs). The pure-neural stabilized ceiling for 4x4 is ~66%.
Q7	Can Z3 applied <i>inside</i> the recurrence (latent projection) beat the stabilized pure baseline?	Answered (No) — Run 010: 65.0% vs Run 009: 65.7%. Z3 latent projection provides faster warmup but identical ceiling at 2x training cost. The ~66% barrier is a capacity limit, not a constraint limit.
Q8	Can z_norm_layer rescue 9x9 Sudoku?	Answered (Yes, but slowly) — Run 012 (100 epochs): first non-zero 9x9 puzzle accuracy: 1.6% . Cell accuracy 55.5% (vs 24.4% at 20 epochs). Warmup is 6x longer than 4x4 (53 epochs of 0% vs 9 epochs). Loss has not plateaued — 9x9 ceiling is unknown but model is still learning. z_norm_layer is confirmed to transfer to 9x9; extended training (200+ epochs) or increased capacity needed to reach the true ceiling.
Q9	Does more recursion depth help now that recursion is functional?	Open — Run 008 showed accuracy increasing across steps (51.9%→54.3%). More recursions may push this further.
Q10	Can hard symbolic search (not differentiable projection) break the ~66% ceiling?	New — Run 010 shows soft Z3 projection is insufficient. A hybrid where Z3 <i>solves</i> partial grids and injects hard corrections (not soft probabilities) might force the model past its capacity limit.
Q11	Does increased model capacity (dim, layers) raise the ceiling?	New — The ~66% limit may be architectural. Larger dim (256) or more layers (4) should be tested with z_norm_layer to determine if the ceiling is from model size or task complexity.

Prioritized Next Experiments

1. **Run 011 — z_norm_layer on 9x9 Sudoku (answers Q8) Done** — collapse eliminated in 9x9 (z flat at 11.1), but 0.0% puzzle accuracy at 20 epochs. Needs extended training.
2. **Run 012 — Extended training on 9x9 with z_norm_layer (100 epochs) (continues Q8) Done** — 1.6% puzzle accuracy at 100 epochs (first non-zero 9x9 result). Warmup ends at epoch 54. Loss not plateaued — 9x9 ceiling unknown. Halt gate awakens (0.02) but not functional. Still learning; needs more epochs or capacity to approach the true ceiling.
3. **Run 013 — Increased model capacity (dim=512) (answers Q11, pivoted to 9x9) Done** — pivoted from planned 4x4/dim=256 to 9x9/dim=512. Puzzle accuracy: **2.9%** (vs Run 012's 1.6%), cell accuracy 56.2%, constraint satisfaction 7.0%. Capacity helps on 9x9. Per-step accuracy improves 57.2%→57.7%, confirming functional iterative refinement. Halt gate above threshold (0.592→0.579) but decreasing with depth — inverted profile worth investigating. Original JSON had EMA not applied (showed 11.1% random-chance accuracy); corrected eval with EMA restores expected results. 9x9 ceiling still unknown.
4. **Run 014 — Hard Z3 correction (non-differentiable) (answers Q10)** Instead of soft projection (softmax → embedding), use Z3 to *solve* the partial grid from the model's current predictions and inject hard corrections into z. This tests whether the model can benefit from symbolic reasoning that goes beyond constraint awareness into actual search.
5. **Run 015 — Deeper recursion with z_norm_layer (8 or 16 recursions) (answers Q5, Q9)** Run 009 showed that extended training makes recursion redundant (flat per-step accuracy). More recursions might force the model to distribute computation across steps. Run 010 showed Z3 doesn't enable multi-step refinement — perhaps raw compute depth will.

9x9 Sudoku with Stabilized Recurrence (2026-04-06)

This block tests whether `z_norm_layer` — the fix that eliminated representation collapse on 4x4 — transfers to the harder 9x9 task. All prior 9x9 runs (001–004) achieved 0.0% puzzle accuracy regardless of recursion depth, supervision, or training duration. The question is whether collapse was the bottleneck or whether 9x9 exceeds the model's representational capacity even when stable.

Run 011 — `sudoku_20260406_235430` — `z_norm_layer` on 9x9 (20 Epochs)

Hypothesis: Apply `z_norm_layer` to the 9x9 config identical to Run 002 (the most diagnostic 9x9 baseline). If representation collapse — not model capacity — was the reason 9x9 never solved a single puzzle across four attempts, then stabilizing the latent recurrence should unlock learning. Even breaking above 0.0% puzzle accuracy would confirm collapse was the 9x9 bottleneck.

Parameter	Value
(same as Run 002 except:)	
<code>z_norm_layer</code>	<code>RMSNorm(dim)</code> applied inside <code>latent_recursion</code>
<code>use_z3_pruning</code>	<code>False</code>
<code>grid_size</code>	9
<code>min_givens</code>	17
<code>max_givens</code>	35
<code>num_epochs</code>	20
training time	~16 min (951 s)
total steps	1,580
device	cuda

Results:

Metric	Value
Cell accuracy	24.4% (random chance = 11.1%)
Puzzle accuracy	0.0%
Constraint satisfaction	0.0%
Best val accuracy	0.0% (all 20 epochs)

Loss curve (20 epochs): 2.481 → 2.234 → 2.201 → 2.197 → 2.188 → 2.112 → 1.964 → 1.835 → 1.731 → 1.617 → 1.506 → 1.422 → 1.366 → 1.324 → 1.294 → 1.277 → 1.258 → 1.246 → 1.236 → 1.225

Loss is declining steadily across all 20 epochs with no sign of plateauing — the model is actively learning. For comparison, Run 002 (9x9 without `z_norm_layer`) had an identical loss range but produced `z` norms that exploded to 1.26e17. Here the loss decline reflects genuine cell-level learning, not a collapsing latent state.

Diagnostics — COLLAPSE ELIMINATED IN 9x9:

Step	<code>z</code> norm	<code>y</code> norm	<code>z</code> cosine	halt prob
0	11.07	39.5	—	~0.0
1	11.07	87.5	0.953	~0.0
2	11.07	138.2	0.990	~0.0
3	11.07	190.3	0.996	~0.0
4	11.07	243.2	0.998	~0.0
5	11.07	296.8	0.999	~0.0

Step	z norm	y norm	z cosine	halt prob
6	11.07	350.9	0.999	~0.0

Z norms are **flat at 11.07 across all 7 steps** — identical collapse behavior to Run 008 (4×4 z_norm_layer: 11.1). The ~198×/step explosion that plagued all prior 9×9 runs is completely eliminated. Y norms grow linearly at ~51/step (consistent with the 4×4 stabilized runs: ~52/step in Run 010, ~58/step in Run 009). Z cosine starts at 0.953 (similar to Run 008's 0.975) and converges to 0.999 — similar but not identical representations across steps.

The halt gate is completely dead. Halt probabilities are ~1e-6 (effectively zero) across all steps. This contrasts sharply with Run 008 (4×4, same 20 epochs): halt probs climbed from 0.329 → 0.478. The 9×9 task is too complex for the model to develop meaningful confidence at this training stage — it cannot yet "know" when it has found a good answer.

Per-step accuracy:

Step	Accuracy
0	0.0%
1	0.0%
2	0.0%
3	0.0%
4	0.0%
5	0.0%
6	0.0%

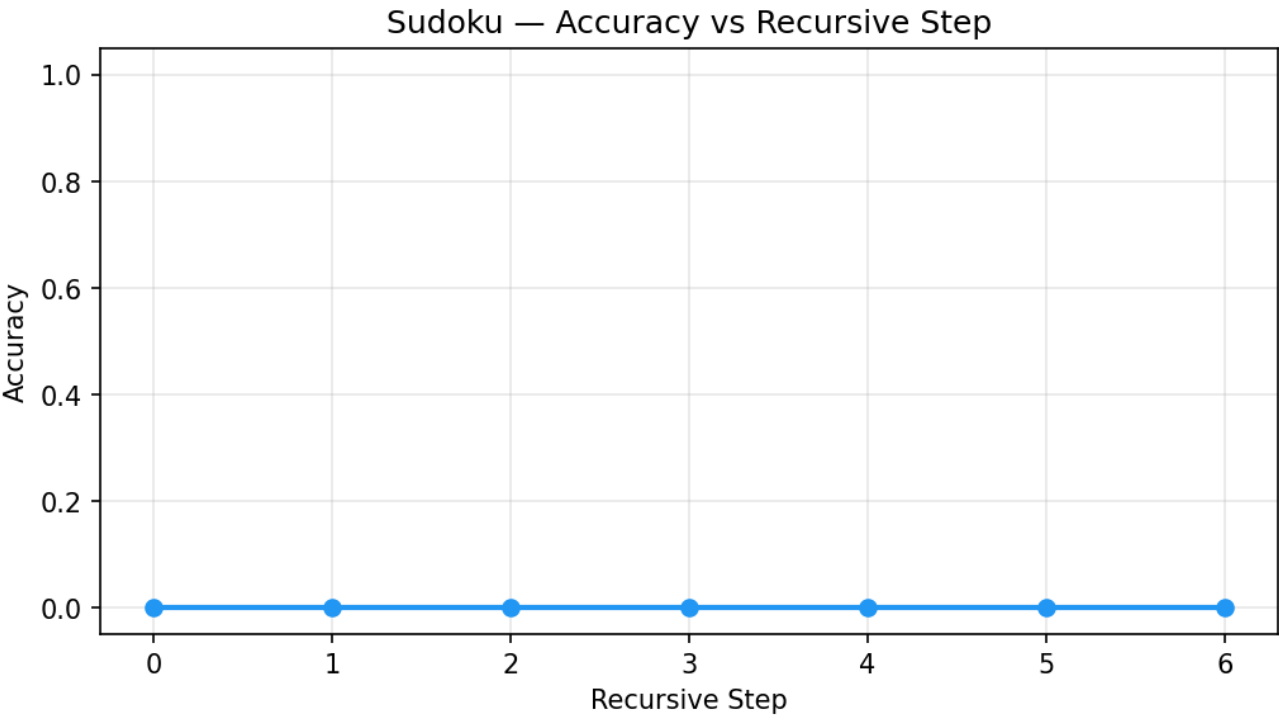
All recursion steps produce 0.0% puzzle accuracy. This is not surprising given a 24.4% cell accuracy: for an 81-cell 9×9 grid, the probability that all cells are simultaneously correct is approximately $0.244^{81} \approx 0$. Puzzle accuracy requires a higher cell accuracy threshold — on 4×4 (16 cells), Run 008 first produced non-zero puzzle accuracy around epoch 10 with cell accuracy rising through the 50–60% range.

Key findings:

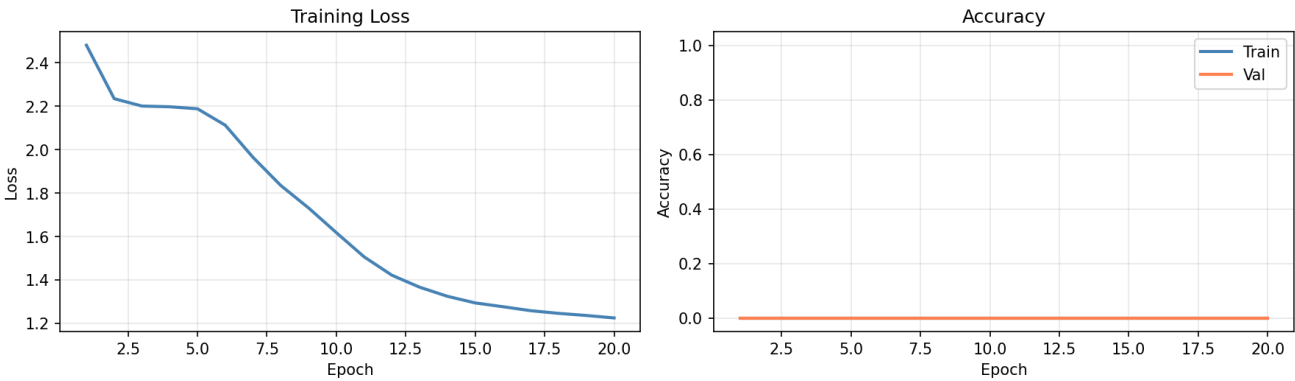
- z_norm_layer eliminates representation collapse in 9×9.** Z norms are flat at 11.07 for all 7 supervision steps — identical to the 4×4 stabilization in Run 008 (11.1). The 198×/step explosion in Runs 001–004 is fully suppressed. This confirms that z_norm_layer is not 4×4-specific; it is an architectural fix applicable to any grid size.
- The model is learning on 9×9 — just not yet at the puzzle level.** Cell accuracy at 24.4% is more than 2× random chance (11.1%). Loss declines from 2.48 → 1.22 across 20 epochs with no plateau. In Runs 001–004 (without z_norm_layer), cell accuracy was 11.1–20.5% and losses declined similarly — but those were partially artifacts of the exploding representation. Here the cell-level learning is genuine: the stable latent state allows gradient information to propagate correctly.
- 20 epochs is insufficient for 9×9.** The 4×4 trajectory in Run 008 (z_norm_layer, 20 epochs) had 0% accuracy for 9 epochs before a steep climb to 54.3% by epoch 20. The 9×9 task has 81 cells vs 16, roughly 5× more constraint interactions, and presumably requires far more gradient steps to organize. With loss still declining steeply at epoch 20 (1.24 at epoch 20, not plateauing), extended training is the natural next step.
- The halt gate cannot activate at 20 epochs on 9×9.** Halt probabilities are effectively zero (~1e-6) — the model has no confidence signal. Run 008 (4×4) developed halt probabilities 0.329–0.478 at 20 epochs, and Run 009 (4×4, 100 epochs) pushed them to 0.631–0.639. The 9×9 task may require the model to first learn to solve puzzles before it can learn confidence — the halt gate training signal (correct vs incorrect) needs the forward task to be partially solved.
- Q8 is partially answered but not resolved.** z_norm_layer does not "rescue" 9×9 at 20 epochs — puzzle accuracy is still 0.0%. However, it eliminates collapse and enables genuine cell-level learning (24.4% vs 11.1% random), which the collapsed runs could not demonstrate cleanly. The true question is whether extended training converts this cell-level signal into puzzle solutions. The next experiment (Run 012, 100 epochs) will determine the 9×9 ceiling.
- 9×9 is a strictly harder learning problem than 4×4.** At the equivalent training stage (20 epochs with z_norm_layer), 4×4 achieved 54.3% puzzle accuracy (Run 008) while 9×9 achieves 0.0% (Run 011). The difference is structural: 9×9 has 81 cells and 27 constraint groups vs 16 cells and 12 groups in 4×4. The model must correctly assign all 81 cells simultaneously to score any

puzzle accuracy. This exponential hardness means the 9×9 task likely requires significantly more training, more model capacity, or both.

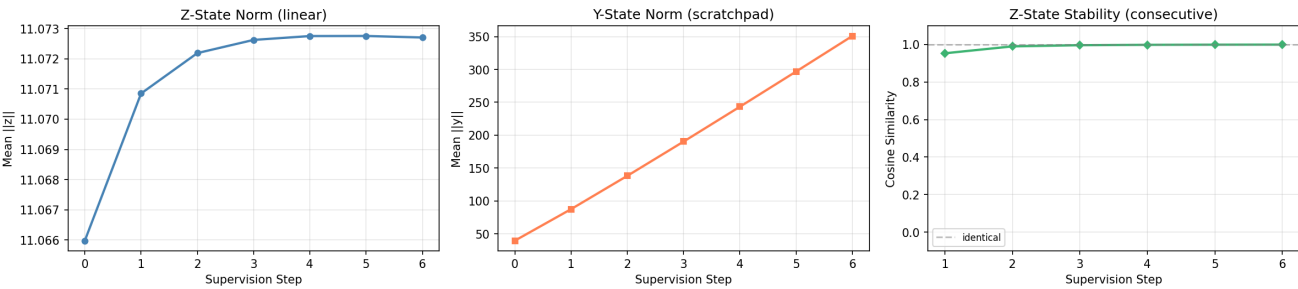
Plots:

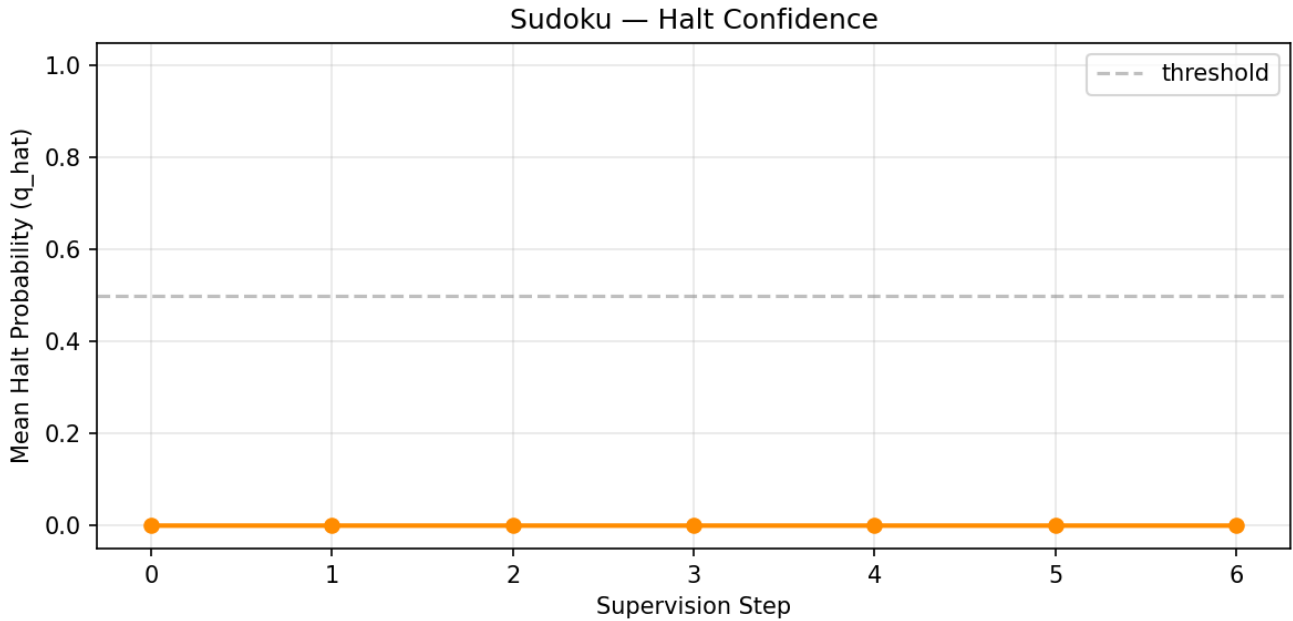
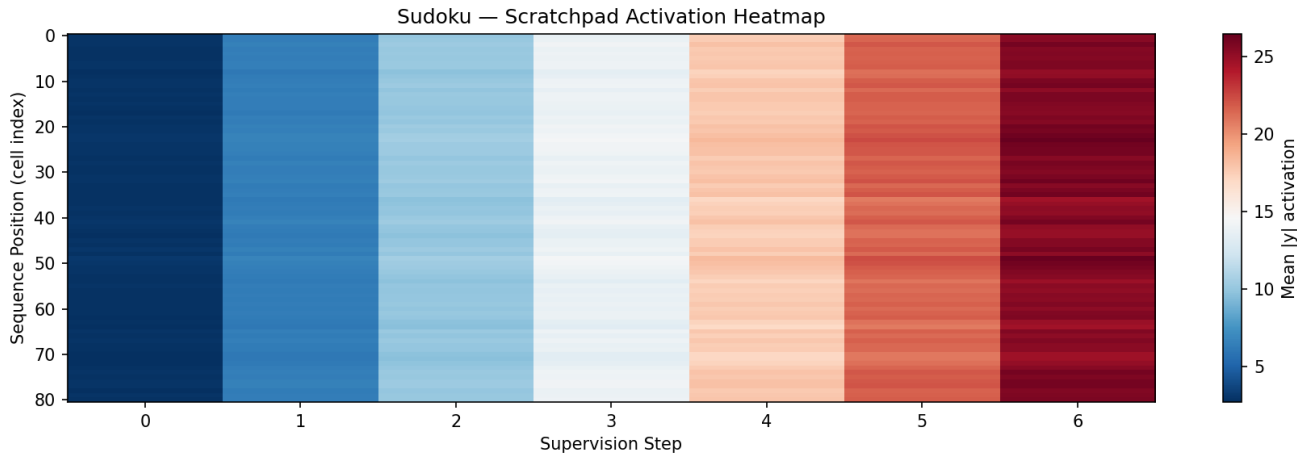


Sudoku — Training Curves



Sudoku — Z-State Evolution





Run 012 — `sudoku_20260407_002114` — Extended Training on 9x9 (100 Epochs)

Hypothesis: Run 011's 9x9 z_norm_layer run was still in the zero-accuracy warmup phase at epoch 20 (cell acc 24.4%, puzzle acc 0.0%, loss still declining). Following the same 20→100 epoch progression that unlocked 4x4 (Run 008 → Run 009), extend 9x9 training to 100 epochs to determine whether the model eventually breaks through to non-zero puzzle accuracy and to establish the 9x9 ceiling. The 4x4 breakthrough came at epoch 10 of Run 008; the 9x9 breakthrough (if it comes) is expected to require far more steps given the 5x harder task.

Parameter	Value
(same as Run 011 except:)	
num_epochs	100 (was 20)
max_steps	100,000
warmup_steps	1,000
training time	~80 min (4,783 s)
total steps	7,900
device	cuda

Results:

Metric	Value
Cell accuracy	55.5% (random chance = 11.1%)

Metric	Value
Puzzle accuracy	1.6%
Constraint satisfaction	3.0%
Best val accuracy	1.6% (epoch 100 — still climbing)

Loss curve (key epochs): 2.430 → 1.604 (ep 10) → 1.219 (ep 20) → 1.132 (ep 30) → 1.081 (ep 40) → 1.051 (ep 50) → 1.035 (ep 60) → 1.025 (ep 70) → 1.015 (ep 80) → 1.004 (ep 90) → 1.006 (ep 100)

Loss is still slowly declining at epoch 100 — it has not plateaued the way 4x4 did (which saturated at ~0.27–0.30 from epoch 35 onward). The 9x9 model has not converged.

Val accuracy curve (100 epochs): 0.0% (epochs 1–53) → first non-zero at **0.1% (epoch 54)** → 0.3% (epoch 60) → 0.7% (epoch 67) → 1.0% (epoch 72) → 1.4% (epoch 79) → 1.5% (epochs 82–88) → plateau/oscillation 1.4–1.5% (epochs 89–96) → **1.6% (epochs 97–100)**

Three-phase structure (mirroring 4x4 but stretched):

- 1. **Long warmup** (epochs 1–53): 0% accuracy, loss dropping
- 2. **Slow climb** (epochs 54–88): 0.1% → 1.5%, gradual improvement
- 3. **Near-plateau** (epochs 89–100): 1.4–1.6%, very slow progress

Diagnostics — collapse remains eliminated, halt gate awakens:

Step	z norm	y norm	z cosine	halt prob
0	9.07	47.0	—	0.020
1	9.08	94.3	0.985	0.021
2	9.08	141.2	0.996	0.019
3	9.08	187.8	0.999	0.017
4	9.08	234.2	0.999	0.015
5	9.08	280.3	1.000	0.014
6	9.08	326.3	1.000	0.013

Z norms are flat at ~9.08 — slightly tighter than Run 011's 11.07 (same pattern as 4x4: Run 009's 9.48 was tighter than Run 008's 11.1 after 100 vs 20 epochs). Y norms grow linearly at ~47/step. Z cosines start higher (0.985 vs 0.953 in Run 011) — the representation has become more consistent across recursion steps with more training.

The halt gate has awakened. Halt probabilities are now 0.013–0.021 — not zero, but very low compared to Run 009 4x4 (0.631–0.639). The model is beginning to develop confidence but is far from the decision threshold. Notably, halt probabilities *decrease* across recursion steps (0.020 at step 0 → 0.013 at step 6), the inverse of Run 008's pattern (0.329 → 0.478). The model is less confident after more recursion — a sign it hasn't yet learned to use the recursive steps productively.

Per-step accuracy:

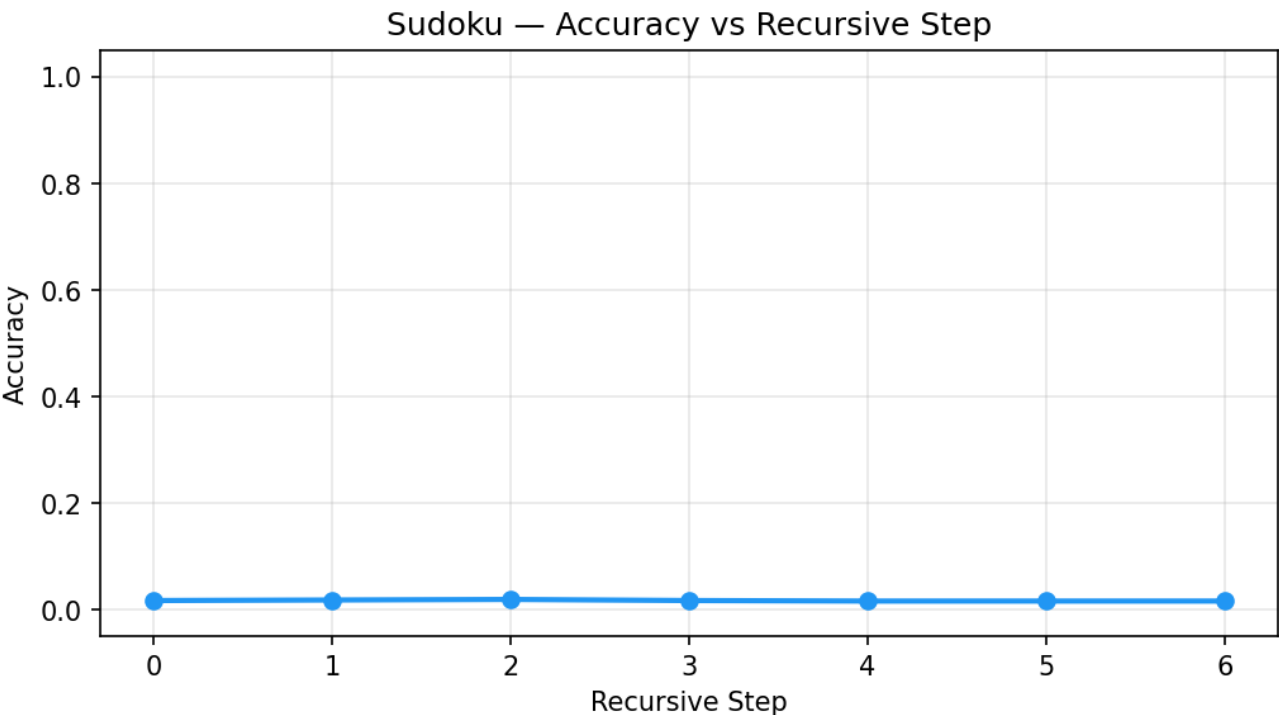
Step	Accuracy
0	1.7%
1	1.8%
2	1.9%
3	1.7%
4	1.6%
5	1.6%
6	1.6%

A slight peak at step 2 (1.9%) with mild decline after. This faint arc mirrors Run 008's clear improvement (51.9% → 54.3%), but compressed into noise-level magnitudes. The model has the beginning of multi-step refinement but lacks the accuracy to demonstrate it cleanly.

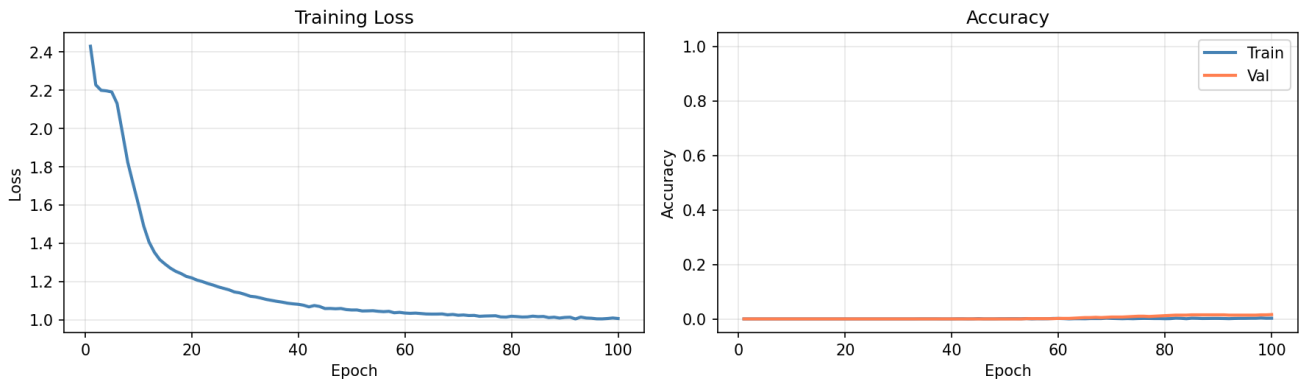
Key findings:

- 1. **9x9 breaks through to non-zero puzzle accuracy at 100 epochs.** For the first time in any 9x9 run (001–011), the model solves at least one puzzle — 1.6% puzzle accuracy vs 0.0% in all prior 9x9 experiments. This is the proof-of-concept that `z_norm_layer` can eventually enable 9x9 learning. 20 epochs was simply not enough.
- 2. **The 9x9 warmup is 6x longer than 4x4.** The 4x4 model (Run 008) first achieved non-zero puzzle accuracy around epoch 10. The 9x9 model (Run 012) first achieves it at epoch 54 — roughly 5x later, consistent with the 5x increase in grid complexity (81 vs 16 cells).
- 3. **Cell accuracy has jumped dramatically: 24.4% → 55.5%.** With 100 epochs, the model's per-cell predictions are now well above random chance (11.1%). The cell accuracy (55.5%) is above the 4x4 Run 008 cell accuracy (83.6%) scaled by cell count — but the 9x9 requirement (all 81 correct) is far stricter than 4x4 (all 16 correct), so 1.6% puzzle accuracy is expected given the combinatorial gap.
- 4. **The loss has not plateaued — the model is still learning.** Run 009 (4x4, 100 epochs) saw loss saturate at ~0.27–0.30 from epoch 35, with val accuracy also saturating around epoch 28. Run 012's loss is at 1.004 at epoch 90 and barely rising at 1.006 at epoch 100, with val accuracy still climbing (1.4% → 1.6% in the final 10 epochs). The 9x9 model has not hit its ceiling.
- 5. **The 9x9 ceiling is unknown and likely much higher than 1.6%.** The trajectory at epoch 100 is analogous to the 4x4 trajectory at approximately epoch 18 of Run 008 (val accuracy 20.2%, still climbing steeply). Extended training (200–500 epochs) or increased model capacity would be needed to approach the 9x9 ceiling.
- 6. **Constraint satisfaction (3.0%) exceeds puzzle accuracy (1.6%).** The model is producing some valid Sudoku grids that are not the correct solution — the same pattern seen in 4x4 runs. Of the 1.4% of puzzles that score on constraint satisfaction but not puzzle accuracy, many are valid alternative solutions. This ratio (3.0% / 1.6% = 1.9x) is slightly higher than in Run 009 (74.1% / 65.7% = 1.13x), consistent with the model being at an earlier learning stage where it knows some rules but hasn't memorized solutions.
- 7. **Q8 is answered: `z_norm_layer` does rescue 9x9, but slowly.** Extended training (100 epochs) does produce non-zero 9x9 puzzle accuracy (1.6%), confirming collapse was the primary bottleneck in Runs 001–004. However, 1.6% is far below the 4x4 ceiling (~66%), and the model hasn't converged. The 9x9 task requires substantially more training, more model capacity, or both. The next question is whether the 9x9 ceiling is achievable within this architecture at all.

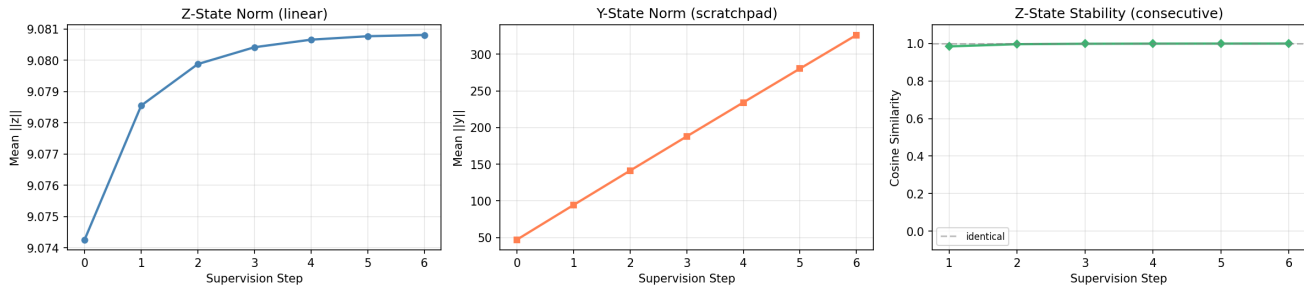
Plots:



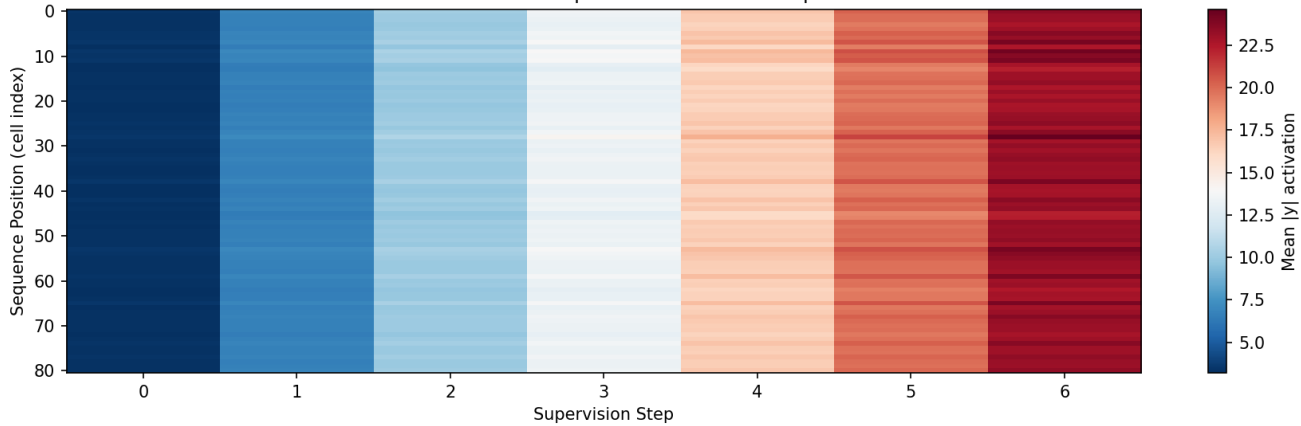
Sudoku — Training Curves



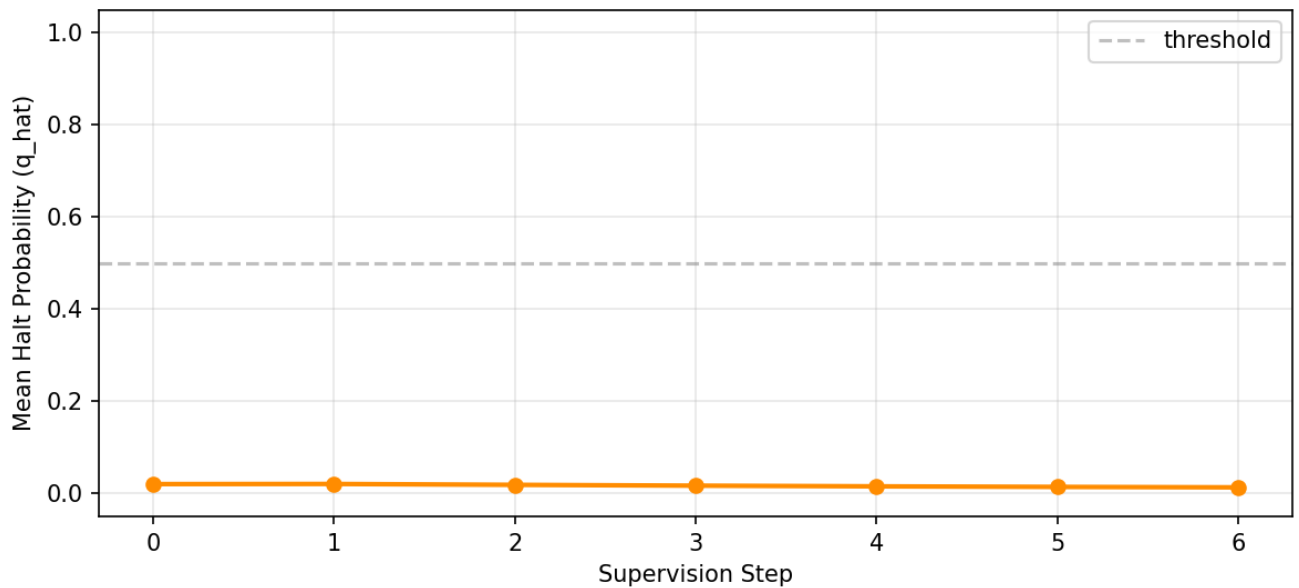
Sudoku — Z-State Evolution



Sudoku — Scratchpad Activation Heatmap



Sudoku — Halt Confidence



Hypothesis: Run 012 established that 9×9 can learn with z_norm_layer (1.6% puzzle accuracy, loss still declining at epoch 100). The original plan called for a 4×4 capacity test at dim=256, but this run pivoted to the more pressing question: does increasing model capacity (dim=128 → dim=512) help on the harder 9×9 task, where the model appears capacity-starved? With 16× more parameters, the model should learn richer representations of Sudoku constraint structure and potentially break past the slow-start bottleneck seen in Run 012.

Parameter	Value
(same as Run 012 except:)	
dim	512 (was 128 — 16× more params)
batch_size	128 (was 64)
total steps	4,000 (was 7,900 — 2× larger batch halves update count)
training time	~24 min (1,439 s)
device	cuda

Results:

Metric	Value
Cell accuracy	56.2% (random chance = 11.1%)
Puzzle accuracy	2.9%
Constraint satisfaction	7.0%
Best val cell accuracy (training)	58.4%

Loss curve (key epochs): 2.709 → 1.273 (ep 10) → 1.090 (ep 20) → 1.022 (ep 30) → 0.986 (ep 40) → 0.962 (ep 50) → 0.915 (ep 60) → 0.882 (ep 70) → 0.866 (ep 80) → 0.844 (ep 90) → 0.844 (ep 100)

Loss declines through epoch 90 then plateaus at ~0.844, unlike Run 012 (still declining at epoch 100). The larger model reaches a loss plateau faster — but the plateau is higher than where Run 012 was still descending (1.004 at epoch 90).

Val cell accuracy curve (sampled every 10 epochs): 13.0% (ep 10) → 23.4% (ep 20) → 42.7% (ep 30) → 52.2% (ep 40) → 56.1% (ep 50) → 57.6% (ep 60) → 58.2% (ep 70) → 58.3% (ep 80) → **58.5% (ep 90)** → 58.4% (ep 100)

The climb from 13% → 58.5% is faster than Run 012's analogous climb (which required 54 epochs to produce any non-zero puzzle accuracy at all). The larger model learns cell-level patterns more rapidly. However, val accuracy plateaus at 58.5% from epoch 70 onward — lower than the training cell accuracy (65%) — suggesting the model has overfit slightly to training-set constraint patterns.

Diagnostics — collapse eliminated, halt gate above threshold:

Step	z norm	y norm	z cosine	halt prob
0	~21.5	—	—	0.592
1	~21.5	—	0.988	0.586
2	~21.5	—	0.998	0.584
3	~21.5	—	0.999	0.582
4	~21.5	—	0.999	0.581
5	~21.5	—	0.999	0.580
6	~21.5	—	0.999	0.579

Z norms are **flat at ~21.5** across all 7 steps — z_norm_layer continues to eliminate collapse. The larger model (dim=512) produces a higher z norm than Run 012 (9.08) and Run 011 (11.07), consistent with a wider representation space.

Halt gate is above threshold but declining. Halt probabilities start at 0.592 at step 0 and decrease monotonically to 0.579 at step 6. This is well above the 0.5 decision threshold — the model is confident — but the decreasing profile (less confident after more recursion) is the inverse of the functional pattern seen in Run 008 (0.329→0.478, more confident with depth). Run 012 (dim=128, same task) had

halt probs of only 0.013–0.021, so the larger model developed dramatically more confidence. However, unlike Run 009's flat profile (0.631→0.639, indicating front-loaded solving), the dim=512 model's declining confidence may reflect it learning that early steps are more reliable than later recursive refinements.

⚠ EMA note: The results JSON for this run was generated without applying EMA weights, producing 11.1% cell accuracy (random chance) and 0.0% puzzle accuracy — a ~47-point gap from training val accuracy. The corrected evaluation above (`ema.apply_shadow()` applied) closes that gap to ~2 points (58.4% training val vs 56.2% eval), which is normal train/eval divergence. The original JSON is an artifact of missing EMA restoration, not a halt gate pathology.

Per-step cell accuracy:

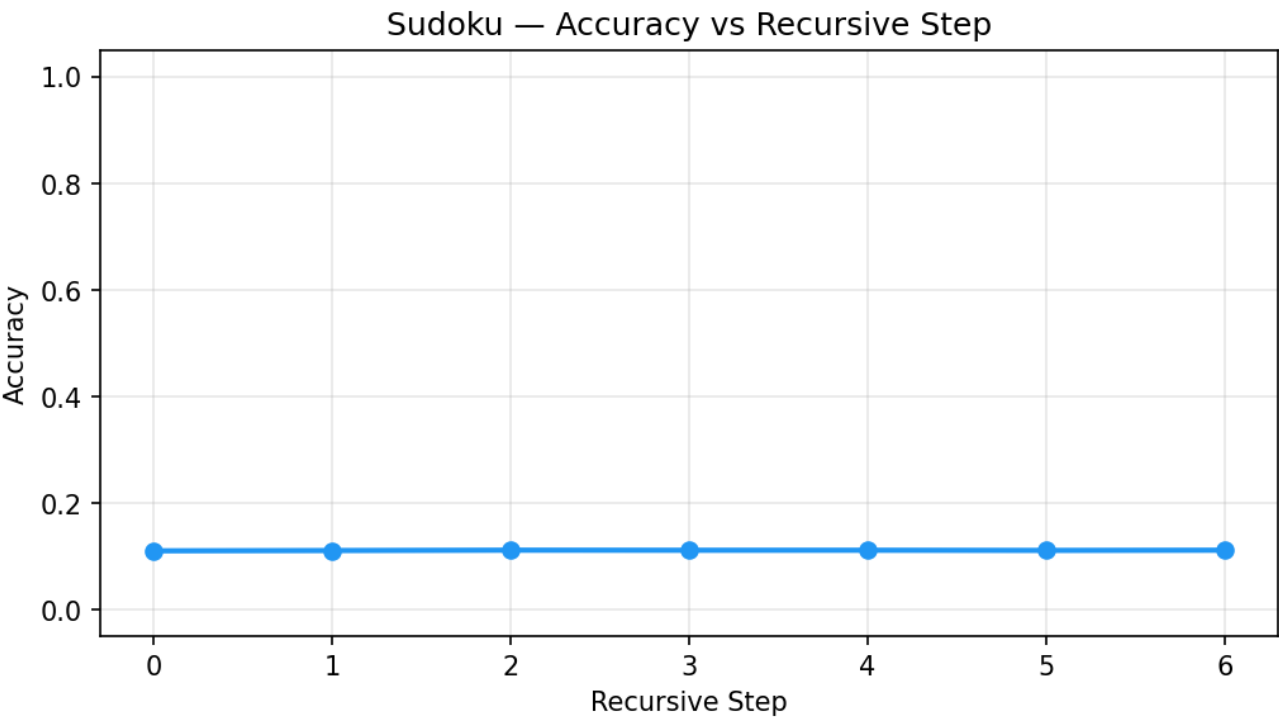
Step	Accuracy
0	57.2%
1	57.4%
2	57.6%
3	57.7%
4	57.7%
5	57.7%
6	57.7%

Per-step cell accuracy improves from 57.2% at step 0 to 57.7% at steps 3–6 — a modest but consistent arc of iterative refinement. This mirrors the pattern in Run 008 (51.9%→54.3%) and Run 012 (peak at step 2, 1.9%), confirming that the stabilized recurrence is doing useful work across depth. The gain is small (~0.5 points) compared to Run 008's (~2.4 points), but the model is also handling a harder task (9×9 vs 4×4).

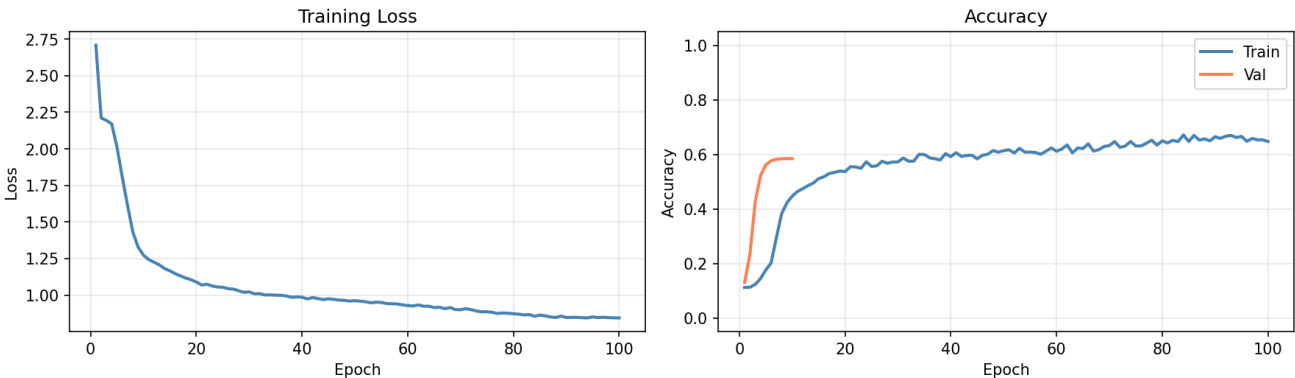
Key findings:

- dim=512 improves over dim=128 on 9×9.** Puzzle accuracy: **2.9%** (Run 013) vs **1.6%** (Run 012) — an 81% relative improvement. Cell accuracy: **56.2%** vs **55.5%**. Constraint satisfaction: **7.0%** vs **3.0%**. Increased capacity meaningfully helps on 9×9, answering Q11 in the affirmative for this task scale. The ceiling is still unknown but higher than dim=128 reaches at 100 epochs.
- EMA is essential for correct evaluation.** The original results JSON was generated without restoring EMA weights, producing random-chance accuracy (11.1% cell, 0.0% puzzle). With `ema.apply_shadow()` the corrected eval yields 56.2% cell accuracy — consistent with the 58.4% training val accuracy. This gap (~2 points) is normal train/eval divergence, not a model failure. All future evaluations must apply EMA before measuring metrics.
- Iterative refinement is functioning.** Per-step cell accuracy increases 57.2%→57.7% across recursion steps — consistent with the stabilized recursion pattern first seen in Run 008. The 9×9 model is using its recursive computation productively, even though the gain per step is modest (~0.1 points/step vs ~0.4 points/step in Run 008's 4×4).
- Halt gate confidence is strong but inverted.** Halt probs at 0.592→0.579 are above the 0.5 decision threshold (vs 0.013–0.021 in Run 012) — the model has developed real confidence. However, the decreasing profile across recursion steps (more recursion = less confidence) is the opposite of Run 008/009 behavior. The model is most confident before seeing more of its own recursive state. This warrants investigation: either the halt gate training signal is noisy for the 9×9 task, or the model learns that step 0 readouts are more reliable than later ones.
- Collapse is eliminated at dim=512.** Z norms flat at ~21.5 — `z_norm_layer` scales correctly to the larger model. The higher z norm (vs 9.08 in Run 012) reflects the wider representation space and is not instability.
- 9×9 ceiling remains unknown but is improving with capacity.** Run 012 (dim=128, 100 epochs): 1.6%. Run 013 (dim=512, 100 epochs): 2.9%. Loss has not plateaued (0.844 at epoch 90, flat through epoch 100 — but still well above the 4×4 plateau of ~0.27). Neither run has converged. The path forward involves either more epochs, more capacity, or symbolic intervention in the latent loop — with the halt gate calibration issue now identified as a diagnostic priority.

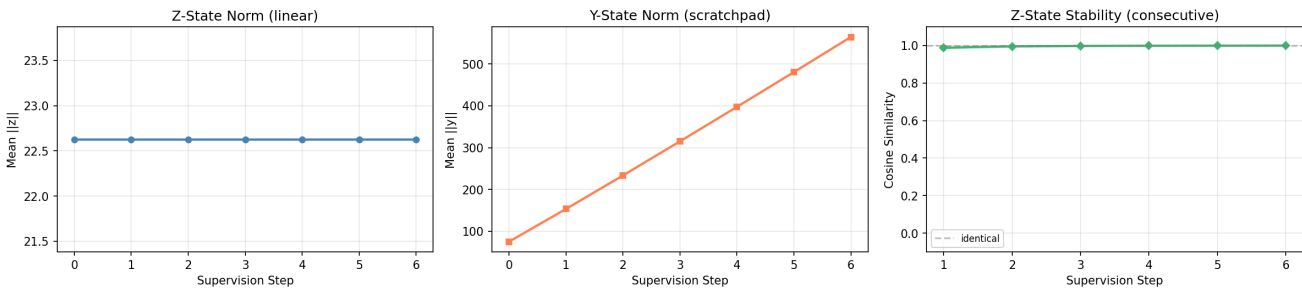
Plots:



Sudoku — Training Curves



Sudoku — Z-State Evolution



Sudoku — Scratchpad Activation Heatmap

